

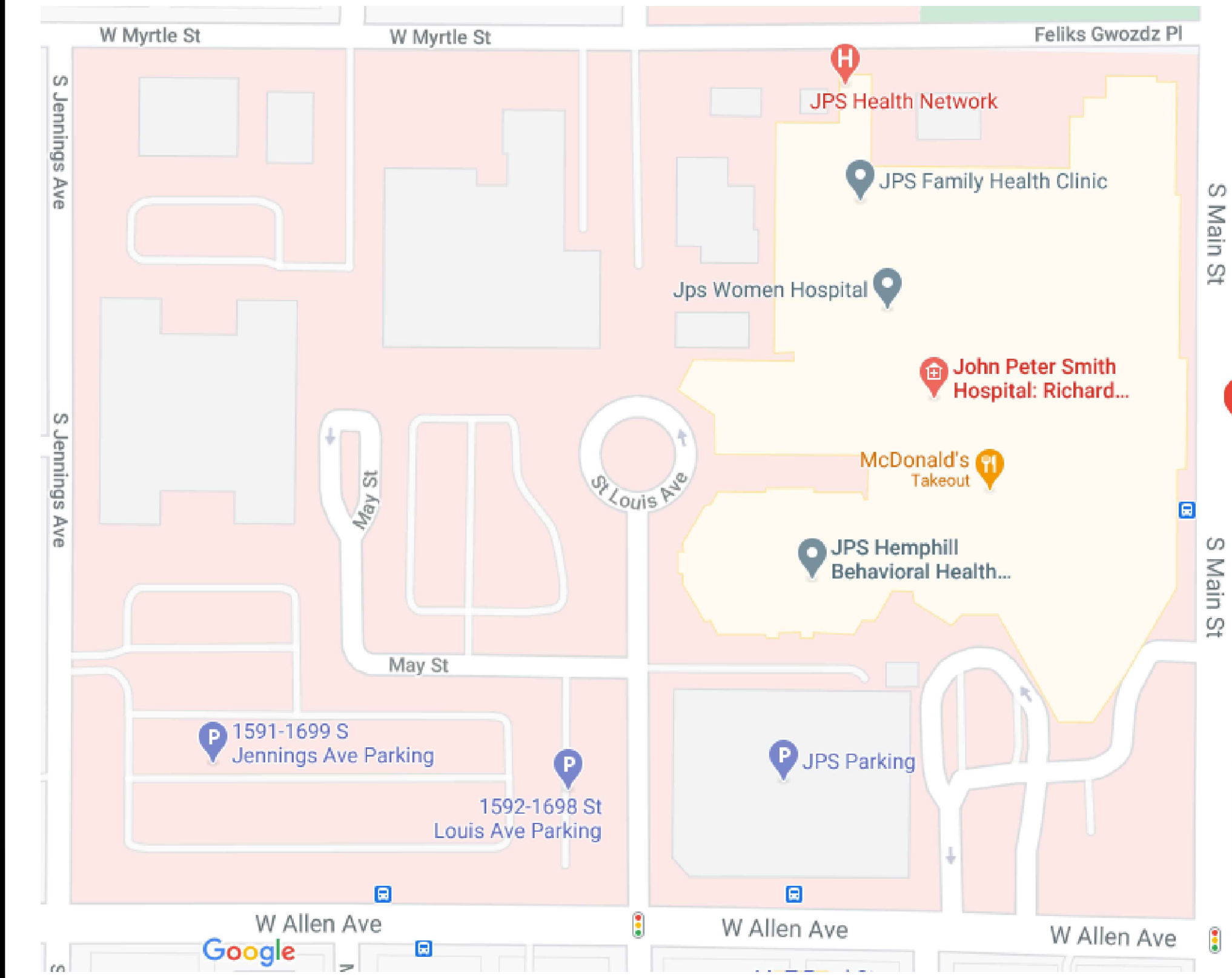
1500 S. Main Street
FORT WORTH, TEXAS 76104

MAIN CENTRAL PLANT
CHILLER ADDITION

FORT WORTH, TX

100% CONSTRUCTION
DOCUMENTS

VINCINITY MAP



SHEET LIST

MEP0.00	COVER SHEET
M0.00	MECHANICAL SYMBOLS & LEGENDS
M1.01 A	MECHANICAL DEMO PLAN
M2.01A	MECHANICAL FLOOR PLAN
M8.01	MECHANICAL CONTROLS
M9.01	MECHANICAL SCHEDULES AND DETAILS
E0.00	ELECTRICAL INFORMATION SHEET
E1.01A	ENLARGED FLOOR PLAN - CHILLER PLANT AREA A - ELECTRICAL DEMOLITION
E1.01B	ENLARGED FLOOR PLAN - CHILLER PLANT AREA B - ELECTRICAL DEMOLITION
E2.01A	ENLARGED FLOOR PLAN - CHILLER PLANT AREA A - ELECTRICAL
E2.01B	ENLARGED FLOOR PLAN - CHILLER PLANT AREA B - ELECTRICAL
E4.00	ELECTRICAL RISER DIAGRAM
E5.00	ELECTRICAL DETAILS

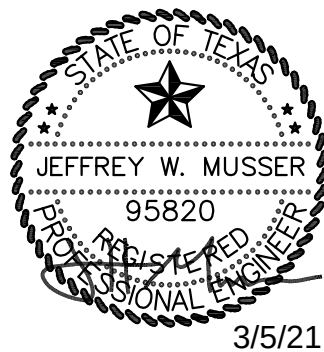
APPLICABLE CODES/STANDARDS

2015 INTERNATIONAL BUILDING CODE WITH LOCAL AMENDMENTS
2015 INTERNATIONAL MECHANICAL CODE WITH LOCAL AMENDMENTS
2015 INTERNATIONAL PLUMBING CODE WITH LOCAL AMENDMENTS
2015 INTERNATIONAL FIRE CODE WITH LOCAL AMENDMENTS
2020 NATIONAL ELECTRICAL CODE WITH LOCAL AMENDMENTS

PROJECT DESCRIPTION

THIS PROJECT CONSISTS OF ADDING A WATER COOLED CHILLER TO THE EXISTING CHILLED WATER SYSTEM IN THE MAIN CENTRAL PLANT OF JPS HOSPITAL LOCATED AT 1500 SOUTH MAIN FORT WORTH, TX.

ISSUE FOR CONSTRUCTION 03-05-21
NO. ISSUED FOR: DATE:



MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE:
PROJECT NO: 121020.00
DRAWN BY:
CHECKED BY:

COVER SHEET

MEP0.00
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1500 S. Main Street
FORT WORTH, TEXAS 76104

HVAC PROJECT NOTES

- GENERAL NOTES FROM THIS LIST SHALL APPLY TO THE ENTIRE MECHANICAL DESIGN.
- ALL WORK SHALL BE PERFORMED IN COMPLIANCE WITH THE LOCAL MECHANICAL CODE, THE LOCAL BUILDING CODES, ADA, OTHER AUTHORITIES HAVING JURISDICTION, AND ENERGY CONSERVATION CODE.
- COORDINATE WITH STRUCTURE AND OTHER DISCIPLINES.
- DRAWINGS ARE SCHEMATIC IN NATURE AND SHALL NOT BE SCALED. THE CONTRACTOR IS RESPONSIBLE FOR CREATING SHOP DRAWINGS AND FOR COORDINATING THE EXACT ROUTING OF ALL DUCTWORK AND PIPING WITH EXISTING FIELD CONDITIONS AND WITH OTHER TRADES.
- PROVIDE VOLUME DAMPER AT SPIN-IN CONNECTION OF DIFFUSER BRANCH INCLUDING THOSE CONNECTING TO THE BOTTOM OF MAIN TRUNK, WHERE VOLUME DAMPER IS ACCESSIBLE. PROVIDE ZONE REGULATORS FOR ALL VOLUME DAMPERS ABOVE INACCESSIBLE CEILING(S) UNLESS OTHERWISE NOTED. COORDINATE CEILING MOUNTED ADJUSTING PORTS WITH EXISTING CEILING ELEMENTS, LINE UP FOR VISUAL PURPOSES. PROVIDE PORT LOCATIONS TO ARCHITECT FOR APPROVAL.
- MAINTAIN MANUFACTURERS RECOMMENDED CLEARANCES ON ALL EQUIPMENT.
- PROVIDE ACCESS CEILING PANELS TO MOTORIZED DAMPERS, VALVES, AND EQUIPMENT ABOVE INACCESSIBLE CEILING(S). COORDINATE LOCATIONS WITH ARCHITECT.
- SEAL ALL DUCT AND PIPE PENETRATIONS THRU EXTERIOR WALLS AIRTIGHT WITH WEATHERPROOF SEALANT.
- FLEX DUCTS SHALL BE 6" MAX. USE RIGID DUCTS IF LENGTH REQUIRED IS MORE THAN 6'-0". FLEX DUCTS SHALL NOT HAVE KINKS THAT WOULD OBSTRUCT AIR FLOW. SIZE FLEX DUCT SUCH THAT PRESSURE DROP IS NOT MORE THAN .08 PER 100'.
- COORDINATE LOCATIONS OF AIR DEVICES WITH LIGHT FIXTURES AND ARCHITECTURAL CEILING DESIGN.
- PRIOR TO ORDERING SHEET METAL VERIFY AND COORDINATE EXACT DUCT ROUTING WITH NEW AND EXISTING STRUCTURAL CONDITIONS.
- CONTRACTOR SHALL VERIFY CLEARANCE REQUIREMENTS AND ROUTING OF NEW SYSTEMS PRIOR TO FABRICATION AS RISES, DROPS, AND OFFSETS MAY BE NECESSARY BECAUSE OF EXISTING FIELD CONDITIONS.
- CONTRACTOR SHALL COORDINATE WITH STRUCTURAL CONDITIONS AND PROVIDE OFFSETS AND CLEARANCES AS REQUIRED.
- INSTALL ALL DUCTWORK AS HIGH AS POSSIBLE OR IN FURR DOWNS WHERE SHOWN.
- COORDINATE FINAL THERMOSTAT AND TEMPERATURE SENSOR LOCATIONS WITH ARCHITECT. DO NOT LOCATE THERMOSTATS OR TEMPERATURE SENSORS BEHIND DOORS OR EQUIPMENT.
- UNLESS SPECIFICALLY INDICATED ALL DUCT, TRANSITIONS SHALL BE SMOOTH AND GRADUAL WITH MAXIMUM DIVERGENT ANGLE OF 15°.
- PROVIDE MINIMUM 6" WIDE FLEXIBLE CONNECTOR TO CONNECT DUCTS TO AIR HANDLING UNITS, FAN COIL UNITS, FAN, ROOF TOP UNITS, AND WHERE INDICATED ON THE DRAWINGS.
- MECHANICAL CONTRACTOR SHALL PROVIDE AND INSTALL ALL WIRING, CONDUIT STARTERS ETC. REQUIRED TO MAKE THE HVAC CONTROL SYSTEM FULLY OPERATIONAL IN ACCORDANCE TO PLANS AND SPECIFICATIONS. WIRING SHALL INCLUDE 120V AND LOW VOLTAGE WIRING REQUIRED FOR HVAC CONTROL SYSTEMS. PROVIDE CONTROLS CAPABLE CONTROLLING MULTI-STAGE HEATING. SUBMIT CONTROL WIRING DIAGRAM TO ENGINEER FOR REVIEW AND APPROVAL PRIOR TO ORDERING OR INSTALLING CONTROL SYSTEM. ALL CONTROL COMPONENTS AND MATERIAL SHALL BE COMPATIBLE WITH THE EQUIPMENT IT SERVES.
- PROVIDE ALL ACCESSORIES FOR LONG LINE SET REFRIGERANT LINES AS REQUIRED BY MANUFACTURER. THIS INCLUDES CRANKCASE HEATER, TXV, HEADSTART ASSIST AND LIQUID LINE SOLENOID. SIZE REFRIGERANT LINES PER MANUFACTURERS RECOMMENDATIONS SUCH THAT LONG LINE SET LOSSES ARE MINIMUM. USE LONG RADIUS ELBOWS AND HAND DRAWN COPPER TUBES. COORDINATE WITH MANUFACTURER.
- ALL DUCT DIMENSIONS SHOWN ON PLANS ARE INTERNAL FREE AREA DIMENSIONS. PROVIDE ACCESS DOORS AND PANELS FOR HVAC ITEMS/COMPONENTS THAT MAY REQUIRED SERVICE ABOVE NON-ACCESSIBLE CEILING(S). COORDINATE ALL PANEL LOCATIONS WITH THE ARCHITECTURAL REFLECTED CEILING PLAN.
- ALL SUPPLY AIR CEILING DEVICES SHALL BE INSULATED ON TOP OF DEVICES TO PREVENT CONDENSATION. INSULATE DEVICES WITH 1-1/2" WRAP AROUND INSULATION AND TOTALLY COVER ALL SURFACES. SECURE INSULATION IN PLACE WITH TAPE. APPLY INSULATION PRIOR TO MOUNTING AIR DEVICES.

LEGEND NOTES

- THIS SHEET IS A GENERAL LIST OF SYMBOLS AND ABBREVIATIONS AND SHALL BE USED AS A DICTIONARY TO DEFINE ITEMS INDICATED ON DRAWINGS. NOT ALL SYMBOLS OR ABBREVIATIONS ARE NECESSARILY USED ON THIS PROJECT.
- THE GENERAL NOTES ON THIS SHEET APPLY TO ALL HVAC DRAWINGS ASSOCIATED WITH THIS PROJECT.

DEMO GENERAL NOTES

- REFER TO MECHANICAL INFORMATION SHEET FOR ABBREVIATIONS AND NOTES.
- DEMOLITION WORK SHALL BE PERFORMED TO ACCOMPLISH REPLACEMENT WORK WITH A MINIMUM AMOUNT OF SYSTEM DOWNTIME.
- SCHEDULE ALL SHUTDOWNS AND DEMOLITION WORK IN ADVANCE WITH OWNER.
- CONTRACTOR SHALL BE RESPONSIBLE TO VERIFY THE CONDITION OF EXISTING EQUIPMENT, EXACT SIZES AND LOCATION OF EXISTING DUCT, PIPING, EQUIPMENT, ETC. BEFORE DEMOLITION WORK BEGINS. REPORT ANY DISCREPANCIES BETWEEN FIELD CONDITIONS AND PLANS TO ARCHITECT AND ENGINEER PRIOR TO THE START OF DEMOLITION WORK.
- CONTRACTOR SHALL REMOVE EXISTING HVAC EQUIPMENT, ASSOCIATED MATERIALS, AND SUPPORTS AS INDICATED. ALL UNUSED EQUIPMENT AND MATERIALS SHALL BE REMOVED BACK TO THE SOURCE. PATCH EXISTING CONDITIONS TO REMAIN AS NECESSARY. COORDINATE WITH OWNER FOR EQUIPMENT TO BE RETURNED TO OWNER. DISPOSE OF ALL UNWANTED EQUIPMENT.
- BALANCING CONTRACTOR SHALL MEASURE AND RECORD EXISTING SYSTEMS AFFECTED BY THIS RENOVATION PRIOR TO THE DEMOLITION WORK. MEASUREMENTS SHALL INCLUDE SUPPLY AND RETURN AIR FLOWS AT EXISTING AIR HANDLING UNITS, SUPPLY, RETURN AND EXHAUST AIR FLOW AT ALL DUCT MAINS, DIFFUSERS AND BRANCHES SERVING EXISTING AREAS TO REMAIN, AND THE FLOW AT PUMPS. BALANCING CONTRACTOR SHALL NOTIFY THE OWNER AND ENGINEER OF ALL DEFICIENCIES FOUND IN ANY OF THE SYSTEMS.
- EXISTING AIR DISTRIBUTION SYSTEMS SHALL BE RE-BALANCED TO MEET THE EXISTING AIR FLOW AS WELL AS THE FLOW REQUIREMENTS INDICATED IN THE CONSTRUCTION DOCUMENTS.

HVAC ABBREVIATIONS

AD	ACCESS DOOR	GALV	GALVANIZED
ADJ	ADJUSTABLE	HP	HORSEPOWER
AFF	ABOVE FINISHED FLOOR	HVAC	HEATING, VENTILATING & AIR CONDITIONING
AFG	ABOVE FINISHED GRADE	ID	INSIDE DIAMETER
ANSI	AMERICAN NATIONAL STANDARDS INSTITUTE	IN	INCHES
ARCH	ARCHITECTURAL	INSUL	INSULATION
ASTM	AMERICAN SOCIETY FOR TESTING MATERIALS	KW	KILOWATT
ASYM	ASYMMETRICAL	LBS	POUNDS
AUX	AUXILIARY	MBH	THOUSAND BRITISH THERMAL UNITS PER HOUR
BFP	BACKFLOW PREVENTER		
BLDG	BUILDING		
BTD	BOTTOM OF DUCT		
BOP	BOTTOM OF PIPE		
BTU/H	BRITISH THERMAL UNIT PER HOUR	MECH	MECHANICAL
COND	CONDENSATE	MEZZ	MEZZANINE
CONT	CONTINUOUS	MFR	MANUFACTURER
CFM	CUBIC FEET PER MINUTE	MTD	MOUNTED
DB	DRY BULB TEMPERATURE	MTR	MOTOR
DA	DIAMETER	NTS	NOT TO SCALE
(E)	EXISTING	NO	NORMALLY OPEN
EXH	EXHAUST	NC	NORMALLY CLOSED
EQMT	EQUIPMENT	O.A.	OUTSIDE AIR
FF	FINISHED FLOOR	OD	OUTSIDE DIAMETER
FLR	FLOOR	ODP	OPEN ENDED DUCT
FT	FEET	OED	OPEN DRIP PROOF
DI	DIGITAL INPUT	SD	SECONDARY DRAIN
DO	DIGITAL OUTPUT	TYP	TYPICAL
DPT	DIFFERENTIAL PRESSURE TRANSMITTER	UL	UNDERWRITERS LABORATORY
FS	FLOW SWITCH	VFD	VARIABLE FREQUENCY DRIVE
CSR	CURRENT SENSING RELAY	VD	VOLUME DAMPER
VFD	VARIABLE SPEED DRIVE	WG	WATER GAUGE

MECHANICAL PIPING SYMBOLS LEGEND

	PIPE UP		CHECK VALVE, SWING GATE
	PIPE DOWN		ANGLE PRESSURE RELIEF VALVE
	CAP		PRESSURE REDUCING VALVE
	90° ELBOW		LOCK SHIELD
	45° ELBOW		QUICK OPENING/CLOSING VALVE
	45° ELBOW DOWN (OGEE)		PRESSURE REGULATOR
	TEE		STRAINER W/BLOW DOWN VALVE
	TEE UP		THREE-WAY VALVE (ELECTRIC)
	TEE DOWN		TWO-WAY VALVE (ELECTRIC)
	TOP CONNECTION		FLEXIBLE CONNECTION
	CROSS		EXPANSION JOINT
	UNION (SCREWED)		THERMOMETER
	UNION (FLANGED)		THERMOMETER WELL
	PIPE BREAK		TEST PLUG
	CONCENTRIC REDUCER		PRESSURE GAUGE W/GAUGE COCK
	ECCENTRIC REDUCER		MANUAL AIR VENT
	END SUCTION PUMP		AUTOMATIC AIR VENT
	BALL VALVE		SOLENOID VALVE
	BUTTERFLY VALVE		FLOW SWITCH
	ISOLATION VALVE		TEMPERATURE AND PRESSURE RELIEF VALVE
	GATE VALVE WITH QUICK DISCONNECT		STEAM TRAP
	TWO-WAY VALVE (PNEUMATIC)		STEAM MOISTURE SEPARATOR
	THREE-WAY VALVE (PNEUMATIC)		STEAM CONTROL VALVE
	BALANCING VALVE		CONTROL, ELECTRIC-PNEUMATIC
	PIPE ALIGNMENT GUIDE		CONTROL, PNEUMATIC-ELECTRIC
	PIPE ANCHOR		RED. PRESS. PRINCIPAL BACKFLOW PREVENTER
	FLANGED END		CHILLED WATER RETURN
	HUMIDIFIER		CHILLED WATER SUPPLY
	HEAT TRACING ON PIPE		CONDENSER WATER RETURN
	DIFFERENTIAL PRESSURE SENSOR		CONDENSER WATER SUPPLY
	NEW CONSTRUCTION		HOT WATER RETURN
	EXISTING TO REMAIN		HOT WATER SUPPLY
	EXISTING TO BE DEMOLISHED		CONDENSATE DRAIN
	CENTER LINE		REFRIGERANT LINES
	DIAMETER		

HVAC NOTES AND SYMBOLS

	CONNECTION POINT OF NEW TO EXISTING
	INDICATES DISCONNECTION POINT
RE: VM-7	REFER TO DETAIL #1 ON DRAWING M-7
MARK-#	EQUIPMENT MARK AND NUMBER. REFER TO EQUIPMENT ABBREVIATIONS ON THIS DRAWING
	AIR DEVICE MARK
	AIR FLOW (CFM)
	NOM. DUCT SIZE (INCHES)
	KEY NOTE DESIGNATIONS
UCD 1"	UNDERCUT DOOR 1"
	INDICATES SECTION NUMBER
	INDICATES DRAWING NUMBER

HVAC NOTES AND DESIGNATIONS

DAMPERS		SENSORS	
	BACKDRAFT DAMPER		H - HUMIDITY SENSOR OR HUMIDISTAT
	FIRE DAMPER AND ACCESS DOOR		T - TEMPERATURE SENSOR OR THERMOSTAT
	MOTORIZED DAMPER		DSZ - DIGITAL ZONE CONTROLLER
	SMOKE DAMPER AND ACCESS DOOR		C - CO2 SENSOR
	FIRE/SMOKE DAMPER AND ACCESS DOOR		SDS - OUTSIDE SENSING DEVICE
	FIRE/SMOKE DAMPER AND ACCESS DOOR		ZS - ZONE SENSOR
	MANUAL VOLUME DAMPER		F - FIRESTAT
			P - PRESSURE MONITOR
			SD - SMOKE DETECTOR

HVAC SHEET LIST

M0.00	MECHANICAL SYMBOLS & LEGENDS
M1.01A	MECHANICAL DEMO PLAN
M2.01A	MECHANICAL FLOOR PLAN
M8.01	MECHANICAL CONTROLS
M9.01	MECHANICAL SCHEDULES AND DETAILS

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MECHANICAL
SYMBOLS AND
LEGENDS

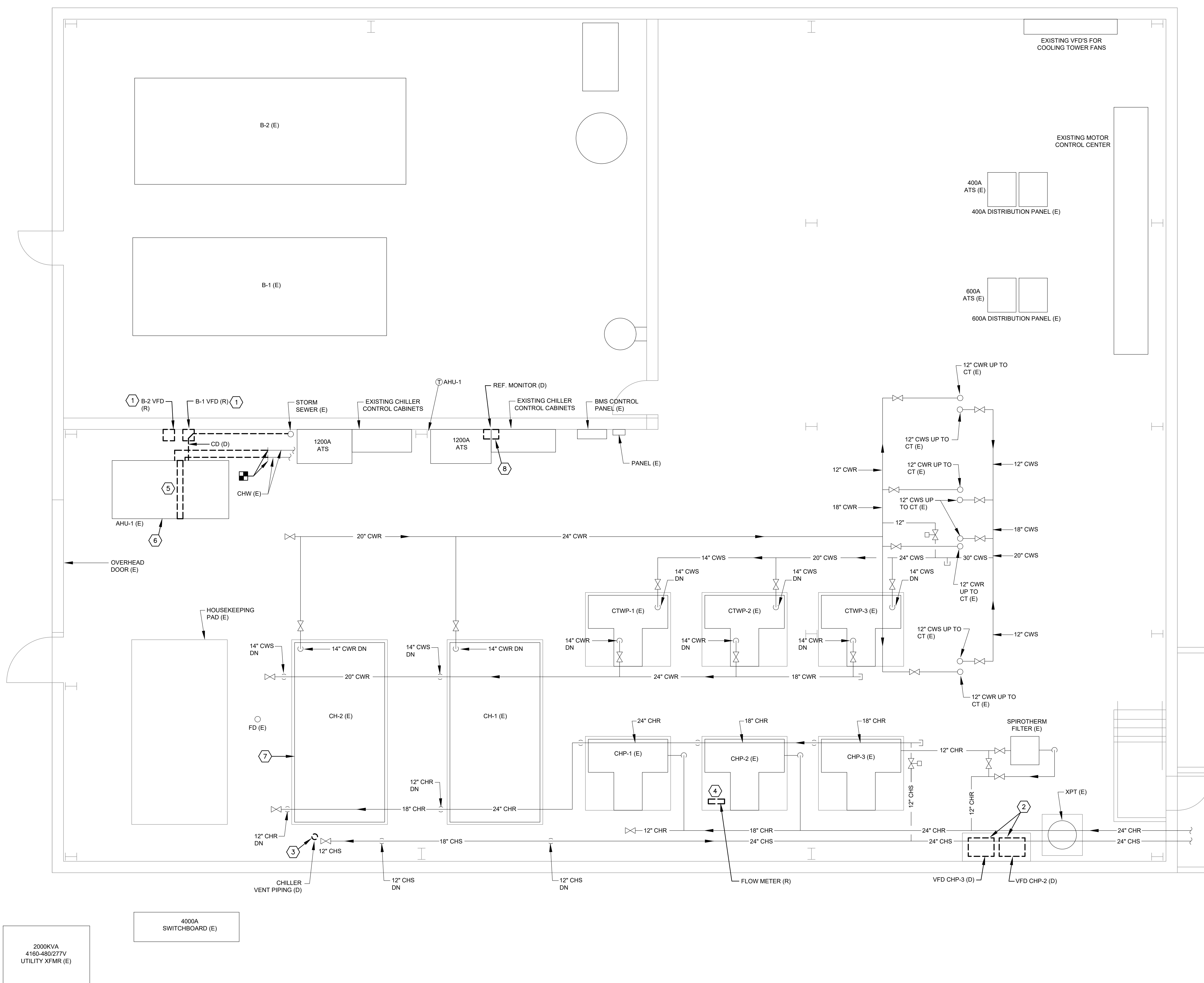
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DEMOLITION GENERAL NOTES

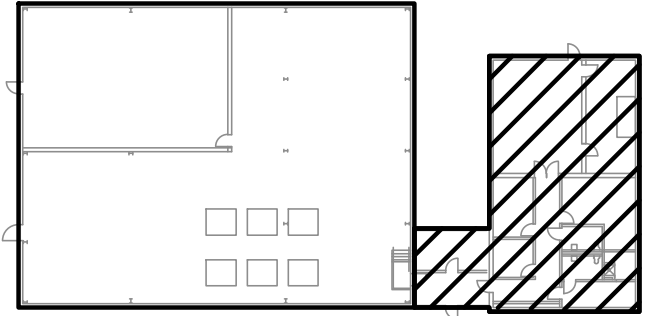
1. REFER TO MECHANICAL INFORMATION SHEET FOR ABBREVIATIONS AND NOTES.
2. DEMOLITION WORK SHALL BE PERFORMED TO ACCOMPLISH REPLACEMENT WORK WITH A MINIMUM AMOUNT OF SYSTEM DOWNTIME.
3. SCHEDULE ALL SHUTDOWNS AND DEMOLITION WORK IN ADVANCE WITH OWNER.
4. CONTRACTOR SHALL BE RESPONSIBLE TO VERIFY THE CONDITION OF EXISTING EQUIPMENT, EXACT SIZES AND LOCATION OF EXISTING DUCT, PIPING, EQUIPMENT, ETC. BEFORE DEMOLITION WORK BEGINS. REPORT ANY DISCREPANCIES BETWEEN PRIOR TO THE START OF DEMOLITION WORK.
5. CONTRACTOR SHALL REMOVE EXISTING HVAC EQUIPMENT, ASSOCIATED MATERIALS, AND SUPPORTS AS INDICATED. ALL UNUSED EQUIPMENT AND MATERIALS SHALL BE REMOVED BACK TO THE SOURCE. PATCH EXISTING CONDITIONS TO REMAIN AS NECESSARY AS DIRECTED BY THE OWNER. CONTRACTOR SHALL TURN OVER TO OWNER OR REMOVE FROM SITE.
6. BALANCING CONTRACTOR SHALL MEASURE AND RECORD EXISTING SYSTEMS AFFECTED BY THIS PROJECT PRIOR TO THE DEMOLITION WORK. MEASUREMENTS SHALL INCLUDE WATER FLOW AT EXISTING CHILLERS, COOLING TOWERS AND PUMPS. BALANCING CONTRACTOR SHALL NOTIFY THE OWNER AND ENGINEER OF ALL DEFICIENCIES FOUND IN ANY OF THE SYSTEMS.
7. EXISTING WATERSIDE DISTRIBUTION SYSTEMS SHALL BE REBALANCED TO MEET THE EXISTING WATER FLOW AS WELL AS THE FLOW REQUIREMENTS INDICATED IN THE CONSTRUCTION DOCUMENTS.

KEYED NOTES

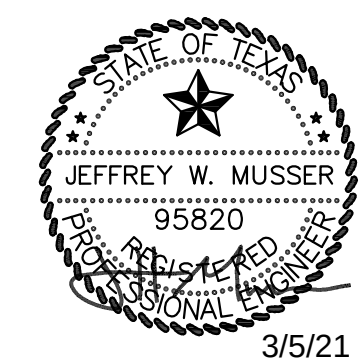
1. RELOCATE EXISTING BOILER VFD. REF. 01M2.01A FOR NEW LOCATION.
2. DEMOLISH EXISTING VFD AND ASSOCIATED HOUSEKEEPING PAD.
3. DEMOLISH EXISTING CHILLER VENT PIPING IN VERTICAL AS REQUIRED TO ACCOMMODATE CONNECTION TO EXISTING CHILLED WATER PIPING MAIN. REF. 01M2.01A.
4. RELOCATE EXISTING FLOW METER AND ASSOCIATED UNISTRUT. REF. 01M2.01A FOR NEW LOCATION.
5. DEMOLISH EXISTING AIR HANDLING UNIT COOLING COIL, ASSOCIATED CONTROL VALVE, MANUAL SHUTOFF VALVES, STRAINER, ETC. TO EXTENT SHOWN.
6. PROVIDE PRE-DEMOLITION AIR AND WATERSIDE TEST AND BALANCE OF EXISTING AIR HANDLER TO ESTABLISH EXISTING AIR AND WATER FLOWS.
7. PROVIDE PRE-DEMOLITION WATER SIDE TEST AND BALANCE ON EXISTING CENTRAL PLANT CONDENSER AND CHILLED WATER SYSTEM S TO ESTABLISH EXISTING WATER FLOWS. SYSTEMS INCLUDE (CHILLERS, COOLING TOWER AND PUMPS).
8. DEMOLISH EXISTING REFRIGERANT MONITORING SYSTEM INCLUDING, SENSORS, HORNS AND STROBES, WIRING AND ASSOCIATED COMPONENTS.



KEY PLAN



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MAIN CENTRAL PLANT
CHILLER ADDITION
SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: JH
CHECKED BY: TF

ENLARGED CHILLER
PLANT AREA A -
MECHANICAL DEMOLITION

M1.01A
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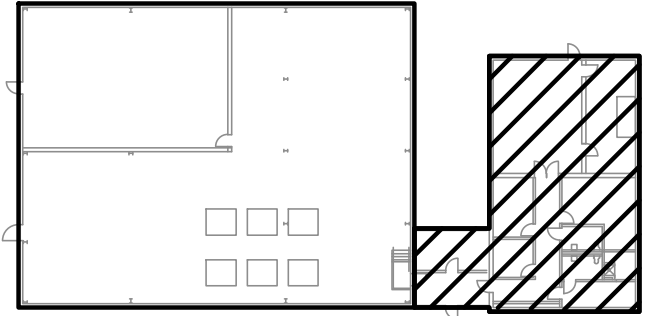
MECHANICAL GENERAL NOTES

1. REFER TO MECHANICAL INFORMATION SHEET FOR ABBREVIATIONS AND NOTES.
2. MAINTAIN MANUFACTURER'S RECOMMENDED CLEARANCES ON ALL EQUIPMENT.
3. CONNECT NEW AND RELOCATED MECHANICAL EQUIPMENT TO EXISTING BUILDING MANAGEMENT SYSTEM.
4. PROVIDE PIPE LABELS FOR NEW PIPING.
5. PROVIDE ENGRAVED LABELS FOR ALL NEW AND RELOCATED MECHANICAL EQUIPMENT INCLUDING CHILLERS, VALVES, VFD'S.
6. REFERENCE COMMISSIONING SPECIFICATION FOR COMMISSIONING REQUIREMENTS AS WELL AS THE CONTRACTOR'S RESPONSIBILITIES DURING THE COMMISSIONING PROCESS.

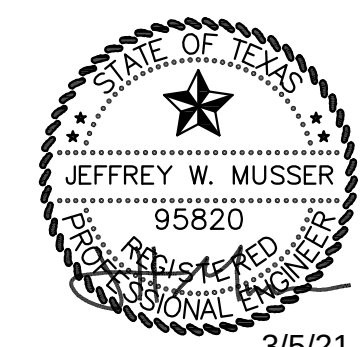
KEYED NOTES

1. RELOCATED VFD. REF. 01/M1.01A FOR PREVIOUS LOCATION. REF. 01/E2.01A FOR ADDITIONAL REQUIREMENTS.
2. PROVIDE 200 HP VFD FOR CHILLED WATER PUMP. MOUNT PUMP ON UNISTRUT.
3. PROVIDE CHILLED WATER PRESSURE DIFFERENTIAL SENSOR(S) MOUNTED 2/3 DOWNSTREAM. COORDINATE FINAL LOCATION IN THE FIELD WITH ENGINEER AND OWNER.
4. CONNECT FULL SIZE CHILLER REFRIGERANT VENT TO CHILLER AT ROUTE UP THROUGH ROOF. FLASH ROOF SLEEVE WATERTIGHT. TERMINATE 12" A.F.R.
5. RE-ROUTE EXISTING CHILLER REFRIGERANT VENT TO AVOID CHILLED WATER PIPING. FIELD VERIFY PIPE SIZE.
6. PROVIDE NEW REFRIGERANT MONITORING SYSTEM INCLUDING MONITOR, HORN AND STROBE, INFRARED SENSORS, AND WIRING. MONITOR SHALL BE COMPATIBLE WITH REFRIGERANT HCFC-123 AND R-514A. SHERLOCK MODEL NO. 404 OR EQUAL.
7. NEW COOLING COIL CC-1. REF. COOLING COIL PIPING DETAIL 03/M8.01 FOR ADDITIONAL REQUIREMENTS.
8. NEW CHILLER SHALL BE TIED INTO EXISTING ALERTON CONTROLS ASCENT COMPASS BUILDING MANAGEMENT SYSTEM. REF. CONTROLS DRAWINGS 01/M8.01 FOR ADDITIONAL REQUIREMENTS.
9. REF. CHILLER DETAIL 01/M8.01 FOR HYDRONIC ACCESSORY REQUIREMENTS.
10. OWNER FURNISHED. CONTRACTOR INSTALLED CHILLER. PROVIDE NEOPRENE VIBRATIONS ISOLATORS.
11. EXISTING HOUSEKEEPING PAD TO BE UTILIZED. EXPAND PAD AS REQUIRED TO ACCOMMODATE NEW CHILLER.
12. REMOTE MOUNTED CHILLER VFD. MAINTAIN MANUFACTURER'S RECOMMENDED AND CODE REQUIRED CLEARANCES.
13. ROUTE CONDENSATE DRAIN LINE DOWN WALL, ACROSS FLOOR AND INTO EXISTING FLOOR DRAIN.
14. RELOCATED FLOW METER AND UNISTRUT. REF. 01/M1.01A FOR PREVIOUS LOCATION.
15. PROVIDE 4" THICK CONCRETE HOUSEKEEPING PAD FOR CHILLER VFD AND ATS. PROVIDE EPOXY COATING. REF. ELECTRICAL DRAWINGS FOR ADDITIONAL ATS INFORMATION.
16. ROUTE CONDENSATE DRAIN LINE TIGHT TO HOUSEKEEPING PAD AS MUCH AS POSSIBLE. SECURE TO FLOOR WITH UNISTRUT CHANNEL BOLTED TO FLOOR TO MATCH EXISTING CONDITIONS.
17. PROVIDE AIRSIDE AND WATERSIDE TEST AND BALANCE BASED ON VALUES ESTABLISHED DURING PRE-DEMOLITION TEST AND BALANCE.
18. PROVIDE WATERSIDE TEST AND BALANCE OF CHILLED AND CONDENSER WATER SYSTEMS.

KEY PLAN



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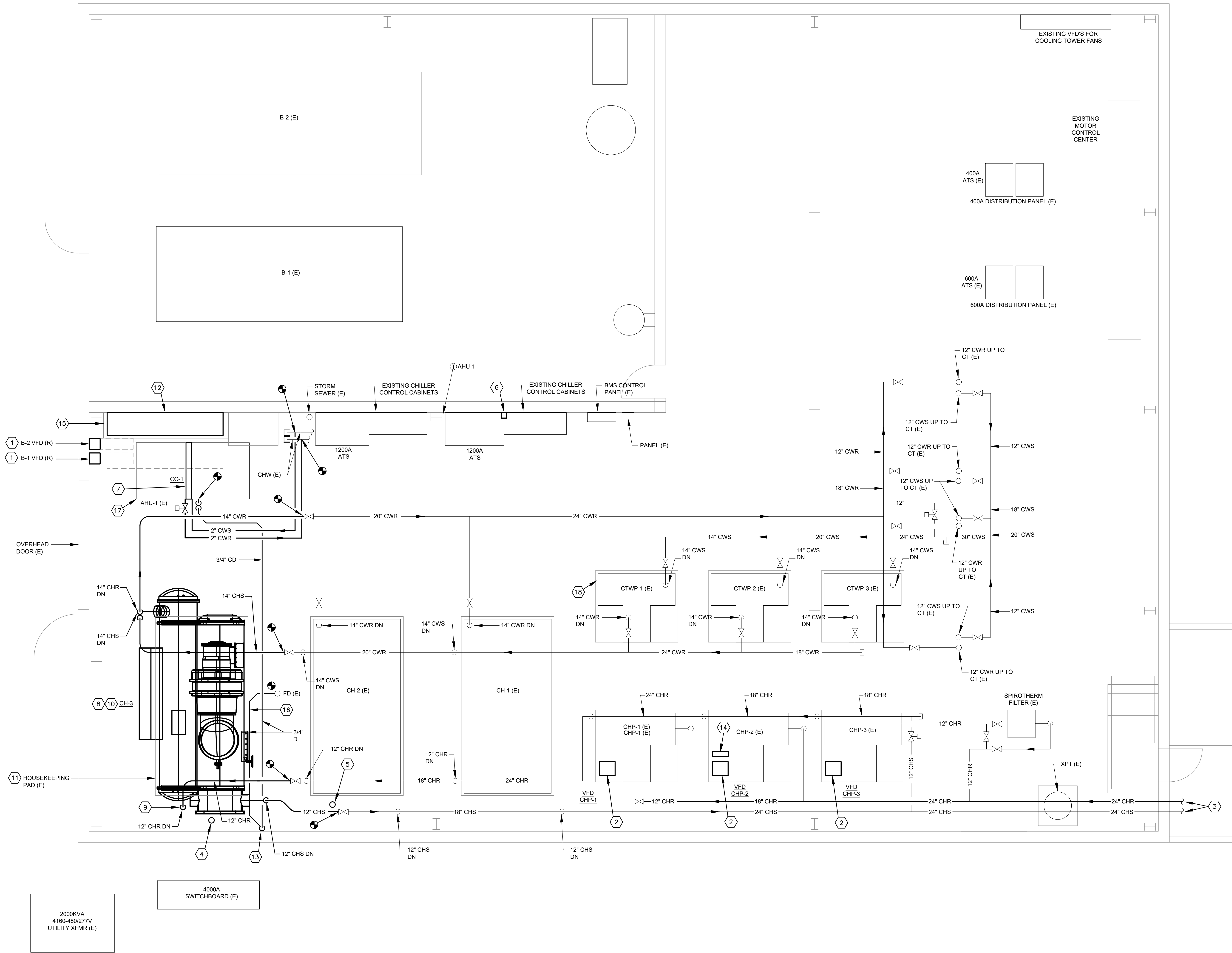


MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: JH
CHECKED BY: TF

ENLARGED CHILLER
PLANT AREA A -
MECHANICAL

M2.01A
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1500 S. Main Street
FORT WORTH, TEXAS 76104

GENERAL NOTES

- THE EXISTING BUILDING MANAGEMENT SYSTEM IS AN ALERTON CONTROLS ASCENT COMPASS SYSTEM. THE NEW MECHANICAL EQUIPMENT SHALL TIE INTO IT AND INTERFACE.
- CONNECT NEW ELECTRICALLY OPERATED CONTROL VALVES TO EXISTING 120V POWER SERVING EXISTING CONTROL VALVES IN THE SAME AREA.
- CONTROLS CONTRACTOR SHALL FIELD VERIFY EXISTING CONDITIONS PRIOR TO BID TO ENSURE EXISTING TO REMAIN CONTROLS COMPONENTS ARE OPERATIONAL AND FULLY FUNCTIONAL TO MEET SEQUENCE OF OPERATION REQUIREMENTS. CONTRACTOR SHALL REPLACE/REPAIR COMPONENTS AS REQUIRED FOR A FULLY FUNCTIONAL SYSTEM.
- EXISTING EQUIPMENT NOT IN SCOPE AND CURRENTLY INTERFACED AND CONTROLLED VIA THE BMS SHALL REMAIN FULLY INTEGRATED.
- ADD NEW CHILLER AND CHILLED WATER PUMP VFD'S AND ALL POINTS TO THE EXISTING BMS FRONT END SYSTEM DISPLAY. FOLLOW EXISTING NAMING CONVENTION FOR ALL VALVES, SENSORS, AND OTHER ADDED COMPONENTS.
- ALL NEW CONTROLS SHALL BE DDC.

GENERAL DESCRIPTION

THE EXISTING BUILDING MANAGEMENT SYSTEM (BMS) SHALL BE UTILIZED TO CONTROL, MONITOR, AND AUTOMATE THE CHILLED AND CONDENSER WATER PRODUCTION PLANT. THE SYSTEM CONSISTS OF (3) CHILLERS, (3) VARIABLE PRIMARY CHILLED WATER PUMPS, AND (3) PRIMARY CONDENSER WATER PUMPS.

THE CHILLED WATER SYSTEM IS A VARIABLE PRIMARY SYSTEM WITH MODULATING BYPASS. THE CONDENSER WATER SYSTEM IS A CONSTANT PRIMARY SYSTEM WITH MODULATING BYPASS.

SEQUENCE OF OPERATION

WHEN THERE IS A CALL FOR CHILLED WATER BY EITHER: OPERATOR COMMAND, AN AIR HANDLING UNIT SEQUENCE, OR OTHER CONTROL EVENTS AS SPECIFIED, THE DDC SYSTEM SHALL PERFORM THE FOLLOWING SEQUENCE:

CHILLED WATER SYSTEM ENABLE/DISABLE

THE CHILLED WATER SYSTEM SHALL BE ENABLED ON A CONTACT CLOSURE FROM ANY SYSTEM CHILLER. WHEN ENABLED, THE BAS CONTROLLER SHALL OPEN THE ISOLATION VALVE AND START THE LEAD CHILLED WATER PUMP. AS ADDITIONAL CHILLERS MAKE CHILLED WATER REQUESTS, THE NEXT PUMP IN THE SEQUENCE SHALL BE ENABLED AND CHILLER ISOLATION VALVES OPENED WHEN THE CHILLED WATER SYSTEM IS DISABLED, THE CHILLED WATER PUMPS SHALL BE OFF AND THE ISOLATION VALVES SHALL BE CLOSED UNLESS REQUESTED BY ONE OF THE CHILLERS. THE ISOLATION VALVE FOR THE FIRST CHILLER IN THE SEQUENCE SHALL BE AN EXCEPTION TO THIS. IF THE PLANT IS DISABLED, THE ISOLATION VALVE FOR THAT CHILLER SHALL BE OPENED SO THAT THE PUMP CAN START IMMEDIATELY WHEN THE PLANT IS ENABLED.

CHILLED WATER PUMP START/STOP

THE BAS CONTROLLER SHALL START A CHILLED WATER PUMP THROUGH A CONTACT CLOSURE OF THE PUMPS VARIABLE FREQUENCY DRIVE (VFD) DRIVE RUN-ENABLE CONTACTS.

CHILLED WATER PUMP STATUS:
THE BAS CONTROLLER SHALL DETECT CHILLED WATER PUMP RUN STATUS BY A VFD CURRENT SWITCH.

CHILLED WATER PUMP LEAD/LAG

THE CHILLED WATER PUMP LEAD/LAG SEQUENCE SHALL BE ROTATED ON A WEEKLY SCHEDULE. THE SEQUENCE SHALL BE BASED ON CALCULATED RUN TIME WITH THE PUMP HAVING THE LEAST RUN TIME AS LEAD. THE PUMP WITH THE HIGHEST RUN TIME SHALL BE THE STANDBY PUMP. FROM THE BAS AN OPERATOR SHALL BE ABLE TO MANUALLY CHANGE THE LEAD/LAG SEQUENCE.

CHILLED WATER PUMP FAILURE

IF THE LEAD START/STOP RELAY IS ENABLED AND THE CURRENT SWITCH STATUS IS OFF FOR MORE THAN 30 SECONDS (ADJ.), THE BAS CONTROLLER SHALL ANNUNCIATE A CHILLED WATER PUMP FAILURE ALARM. TO THE BAS AND START THE STANDBY PUMP. ONCE THE PROBLEM HAS BEEN CORRECTED, THE OPERATOR SHALL BE ABLE TO CLEAR THE ALARM FAILURE FROM THE BAS CONTROLLER USING TRACER TU, FROM A BAS OR BY MANUALLY OVERRIDING THE PUMP ON MOMENTARILY. THIS SHALL RE-ENABLE THE LEAD/LAG SEQUENCE.

CHILLED WATER PUMP SPEED

THE BAS CONTROLLER SHALL MONITOR THE CHILLED WATER SYSTEM DIFFERENTIAL PRESSURE SENSOR. WHEN THE PUMPS VFD IS ENABLED, THE BAS CONTROLLER SHALL CONTROL THE ANALOG SPEED SIGNAL THAT IS SENT TO THE PUMPS VFD TO MAINTAIN THE CHILLED WATER DIFFERENTIAL PRESSURE SETPOINT OF 15 PSIG (ADJ.).

CHILLER ISOLATION VALVES

CHILLER ISOLATION VALVES SHALL PREVENT THE FLOW OF WATER THROUGH NON-OPERATING CHILLERS. CHILLER CHILLED WATER PUMP OPERATION WILL BE COORDINATED WITH THE ISOLATION VALVE OPERATION.

CHILLED WATER BYPASS VALVE CONTROL

THE BAS CONTROLLER SHALL MONITOR THE EVAPORATOR DIFFERENTIAL PRESSURE OF THE CHILLER. WHEN THE PRESSURE OF THE OPERATING CHILLER(S) INDICATES A LOW PRESSURE (FLOW), THE BAS CONTROLLER SHALL CONTROL THE ANALOG SIGNAL THAT IS SENT TO THE CHILLED WATER BYPASS VALVE TO MAINTAIN THE MINIMUM PRESSURE (FLOW) SETPOINT (ADJ.) OF THE CHILLERS.

ADDING A CHILLER

THE BAS SHALL MONITOR THE CHILLED WATER SUPPLY AND RETURN TEMPERATURE AND THE CHILLED WATER FLOW THROUGH THE LEAD CHILLER. A CHILLER WILL BE ADDED IF ANY OF THE FOLLOWING OCCUR:
- CHILLED WATER SUPPLY TEMPERATURE AS SENSED BY IS GREATER THAN SETPOINT PLUS OFFSET FOR AN ADJUSTABLE TIME PERIOD
- THE MAXIMUM EVAPORATOR FLOW OF THE LEAD CHILLER IS REACHED, AND THE BYPASS VALVE IS CLOSED FULLY, FOR AN ADJUSTABLE TIME PERIOD

CONDENSER WATER SYSTEM ENABLE/DISABLE

THE CONDENSER WATER SYSTEM SHALL BE ENABLED ON A CONTACT CLOSURE FROM ANY CHILLER. WHEN ENABLED, THE BAS CONTROLLER SHALL START THE LEAD CONDENSER WATER PUMP AND CONTROL THE COOLING TOWER FANS AND TOWER BYPASS VALVE TO MAINTAIN THE CONDENSER WATER SETPOINT (ADJ.). AS ADDITIONAL CHILLERS MAKE CONDENSER WATER REQUESTS, THE LAG PUMP(S) WILL BE ENABLED. WHEN THE CONDENSER WATER SYSTEM IS DISABLED, THE CONDENSER WATER PUMP AND COOLING TOWER FAN SHALL BE OFF.

CONDENSER PUMP START/STOP

THE CONDENSER WATER PUMPS WILL BE CONTROLLED THROUGH THE ON-OFF SELECTOR SWITCH ON THE ASSOCIATE MOTOR STARTERS OR VIA THE BMS AS FOLLOWS:

HAND - THE PUMP WILL RUN CONTINUOUS
OFF - THE PUMP WILL BE OFF

CONDENSER PUMP STATUS

THE BAS CONTROLLER SHALL DETECT CONDENSER WATER PUMP RUN STATUS BY A CURRENT SENSING RELAY.

CONDENSER PUMP LEAD/STANDBY

THE CONDENSER PUMP LEAD/LAG SEQUENCE SHALL BE ROTATED ON A WEEKLY SCHEDULE. THE SEQUENCE SHALL BE BASED ON CALCULATED RUN TIME WITH THE PUMP HAVING THE LEAST RUN TIME AS LEAD. THE PUMP WITH THE NEXT LOWEST RUN TIME WILL BE THE SECOND IN THE SEQUENCE AND SO ON FROM THE BAS OR WITH TRACER TU AN OPERATOR SHALL BE ABLE TO MANUALLY CHANGE THE LEAD/LAG SEQUENCE.

CONDENSER PUMP FAILURE

IF THE LEAD START/STOP RELAY IS ENABLED AND THE CURRENT SWITCH STATUS IS OFF FOR MORE THAN 30 SECONDS (ADJ.), THE BAS CONTROLLER SHALL ANNUNCIATE A CONDENSER PUMP FAILURE ALARM TO THE BAS AND SHALL START THE NEXT PUMP IN THE SEQUENCE. ONCE THE PROBLEM HAS BEEN CORRECTED, THE OPERATOR SHALL BE ABLE TO CLEAR THE ALARM FAILURE FROM THE BAS CONTROLLER USING TRACER TU, FROM A BAS OR BY MANUALLY OVERRIDING THE PUMP ON MOMENTARILY. THIS SHALL RE-ENABLE THE LEAD/LAG/STANDBY SEQUENCE.

CONDENSER PUMP SPEED

THE CONDENSER WATER PUMPS WILL OPERATE AT CONSTANT SPEED.

CHILLER ISOLATION VALVES

CHILLER ISOLATION VALVES SHALL PREVENT THE FLOW OF CONDENSER WATER THROUGH NON-OPERATING CHILLERS. CHILLER CONDENSER WATER PUMP OPERATION SHALL BE COORDINATED WITH THE ISOLATION VALVE OPERATION.

TOWER ISOLATION VALVES

COOLING TOWERS SHALL OPERATE WITH CONDENSER WATER PUMPS. TOWER ISOLATION VALVES SHALL CONTROL IN CONJUNCTION WITH THE CONDENSER WATER SYSTEM AND ALLOW TOWER TEMPERATURE CONTROL TO FUNCTION.

TOWER BYPASS VALVE CONTROL

WHEN THE CONDENSER WATER SYSTEM IS ENABLED, THE COOLING TOWER SHALL MAINTAIN THE ENTERING CONDENSER WATER TEMPERATURE BY FIRST MODULATING THE NORMALLY OPEN BYPASS VALVE. WHEN THE VALVE IS IN FULL TOWER BYPASS AND THE ENTERING CONDENSER WATER TEMPERATURE IS GREATER THAN THE ENTERING CONDENSER WATER TEMPERATURE SETPOINT MINUS 2.0 DEG. F (ADJ.), THE BYPASS VALVE SHALL BEGIN TO CLOSE SENDING WATER TO THE COOLING TOWER. THE BYPASS VALVE SHALL CONTINUE TO MODULATE CLOSED TO MAINTAIN THE ENTERING CONDENSER WATER TEMPERATURE SETPOINT WITHOUT THE TOWER FANS WHEN THE BYPASS VALVE IS FULLY CLOSED AND THE ENTERING CONDENSER WATER TEMPERATURE REACHES CONDENSER WATER TEMPERATURE SETPOINT PLUS 2.0 DEG. F (ADJ.) THE CONDENSER WATER TEMPERATURE SHALL BE MAINTAINED BY OPERATING THE COOLING TOWER FANS TO MAINTAIN THE ENTERING CONDENSER WATER TEMPERATURE SETPOINT.

TOWER FAN START/STOP

THE BAS CONTROLLER SHALL START A TOWER FAN THROUGH A CONTACT CLOSURE OF THE FAN VFD RUN-ENABLE CONTACTS.

TOWER FAN STATUS

THE BAS CONTROLLER SHALL DETECT TOWER FAN STATUS BY A VFD CURRENT SWITCH.

TOWER FAN CONTROL

COOLING TOWER SEQUENCE SHALL BE ROTATED ON A WEEKLY SCHEDULE. FROM THE BAS OR THE SERVICE TOOL AN OPERATOR SHALL BE ABLE TO MANUALLY CHANGE THE STAGING SEQUENCE. WHEN A CHILLER IS OPERATING AND THE COOLING TOWER BASIN TEMPERATURE RISES TO 2.0 DEG. F (ADJ.) ABOVE THE CURRENT TOWER LEAVING WATER SETPOINT THE LEAD COOLING TOWER FAN SHALL BE TURNED ON AT MINIMUM SPEED AND THE DDC CONTROL LOOP ENABLED. THE SYSTEM SHALL THEN DELAY ADDING ADDITIONAL TOWERS FOR 3 MINUTES (ADJ.). WHEN THE OPERATING FANS ARE OPERATING AT 50 PERCENT SPEED (ADJ.), AN ADDITIONAL TOWER SHALL BE ENABLED AND CONTROLLED AT THE SAME SPEED AS THE OPERATING FANS UNTIL ALL ACTIVE COOLING TOWERS ARE ENABLED. WHEN OPERATING FANS ARE RUNNING AT MINIMUM SPEED AND THE TOWER SUPPLY WATER TEMPERATURE IS 5.0 DEG. F (ADJ.) BELOW THE CURRENT TOWER LEAVING WATER SETPOINT THE MOST LAG TOWER FAN SHALL BE TURNED OFF. THE SYSTEM SHALL THEN DELAY SUBTRACTING ADDITIONAL FANS FOR 3 MINUTES (ADJ.). COOLING TOWER FANS SHALL HAVE 5 MINUTE (ADJ.) MINIMUM ON AND OFF DELAYS.

TOWER FAN FAILURE

IF THE LEAD TOWER FAN START/STOP RELAY IS ENABLED AND THE CURRENT SWITCH STATUS IS OFF FOR MORE THAN 15 SECONDS (ADJ.), THE BAS CONTROLLER SHALL ANNUNCIATE A TOWER FAN FAILURE ALARM TO THE BAS AND START THE NEXT TOWER CELL IN THE SEQUENCE. WHEN A TOWER FAN FAILURE EXISTS, LEAD/LAG AUTOMATION SHALL BE DISABLED AND THE CURRENTLY RUNNING TOWER CELL BECOMES LEAD CELL. ONCE THE PROBLEM HAS BEEN CORRECTED, THE OPERATOR SHALL BE ABLE TO CLEAR THE ALARM FAILURE FROM THE BAS CONTROLLER USING TRACER TU, FROM A BAS, OR BY MOMENTARILY OVERRIDING THE FAN ON. THIS ACTION SHALL RE-ENABLE THE LEAD/LAG SEQUENCE.

COOLING TOWER SUMP CONTROL

THE SUMP HEAT SHALL ONLY BE ENABLED WHEN THE OUTDOOR TEMPERATURE IS BELOW 38.0 DEG. F (ADJ.) AND THE SYSTEM IS DISABLED.

SUMP HEATER: ALL COOLING TOWERS SHALL HAVE A SUMP TEMPERATURE SENSOR AND SUMP HEATER. COOLING TOWER SUMP HEATERS SHALL BE CONTROLLED TO MAINTAIN A SUMP TEMPERATURE OF 45.0 DEG. F (ADJ.) WITH THE FOLLOWING SCHEDULE.

OUTSIDE AIR TEMPERATURE CONTROLLED

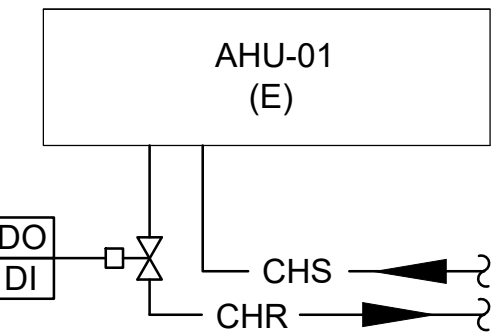
45.0 DEG. F (ADJ.) ON
48.0 DEG. F (ADJ.) OFF

02 CHILLED/CONDENSER WATER SEQUENCE OF OPERATION

NO SCALE

NOTE:

AHU-01 SEQUENCE OF OPERATION SHALL REMAIN AS CURRENTLY PROGRAMMED.

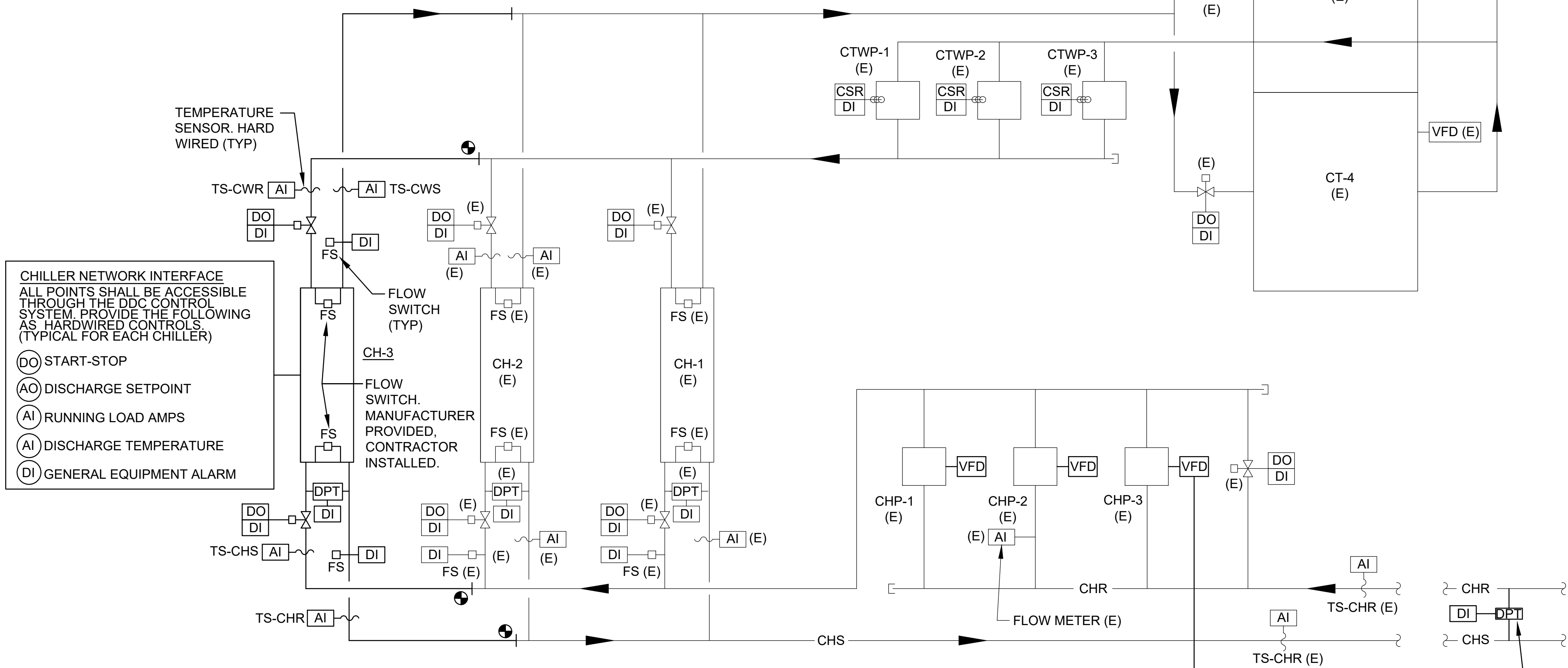


03 AHU-01 CHILLED CONTROL DIAGRAM

NO SCALE

01 CHILLED/CONDENSER WATER CONTROL DIAGRAM

NO SCALE



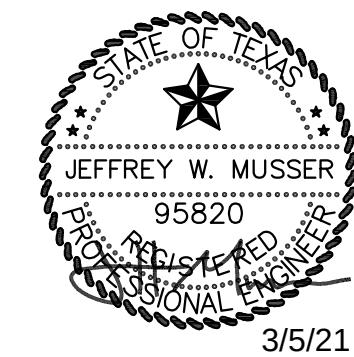
VFD NETWORK INTERFACE

ALL POINTS SHALL BE ACCESSIBLE THROUGH THE DDC CONTROL SYSTEM. PROVIDE THE FOLLOWING AS HARDWIRED. TYPICAL FOR ALL VFD'S

- (DO)-(RF-RUN) START-STOP
- (AO)-(RF-SPD) SPEED CONTROL (% FULL)
- (DI)-(RF-ALM) GENERAL VFD ALARM
- (DI)-(RF-STS) FAN STATUS
- (AI)-(RF-FBK) VFD FEEDBACK

COORDINATE LOCATION AND QUANTITY OF DP SENSORS REQUIRED IN FIELD WITH FACILITIES ENGINEERING

ISSUE FOR CONSTRUCTION 03-05-21
NO. ISSUED FOR: DATE:



MAIN CENTRAL PLANT
CHILLER ADDITION

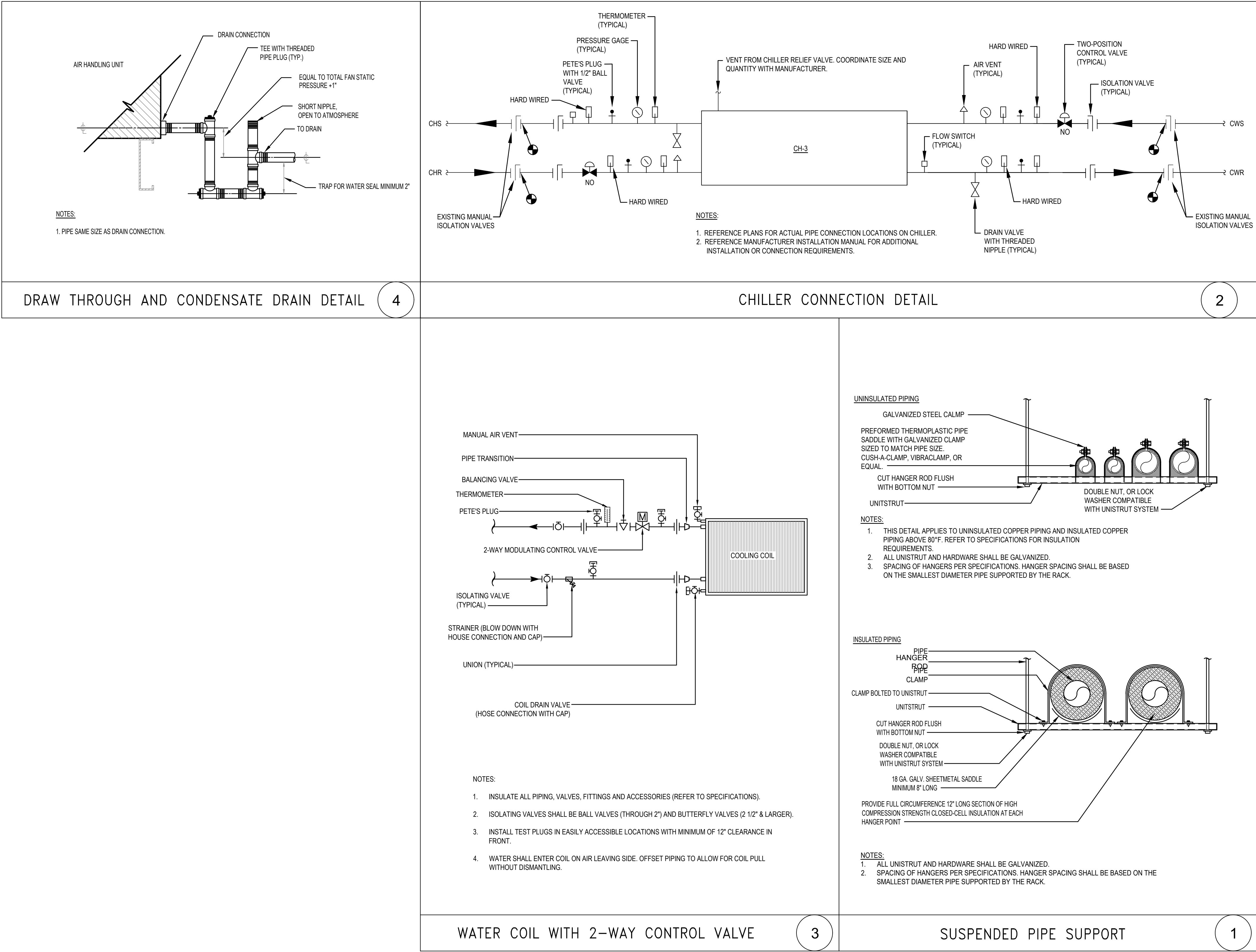
SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: JH
CHECKED BY: TF

MECHANICAL
CONTROLS

M8.01
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COOLING COIL SCHEDULE															
MARK	SERVES	TYPE	CFM	MAX FACE VELOCITY (FPM)	MIN #ROWS/ FINS/FT.	EAT DBWB (°F)	LAT DBWB (°F)	EWT/LWT (°F)	SENS CAPACITY (MBH)	TOTAL CAPACITY (MBH)	GPM	MAX WTR. PRESSURE DROP (F.T.)	MAX AIR PRESSURE DROP (F.T.)	MANUFACTURER/MODEL NO.	NOTES
CC-1	AHU-01	CHILLED WATER	8,800	525	4/72	80.0/67.0	63.1/60.2	42/52	165.6	195.8	39.0	2.8	0.4	TRANE/DUM837065G0D8072CAA008	ALL
NOTES: 1. CONTRACTOR TO COORDINATE COIL CONNECTION SIZE WITH THE MECHANICAL DRAWINGS AND ACTUAL FIELD CONDITIONS. 2. 0.16" WALL THICKNESS. 3. COPPER TUBES/ALUMINUM FINS.															

WATER COOLED CHILLER SCHEDULE (FOR REFERENCE ONLY)																		
MARK	TYPE	NET CAPACITY (TONS)	EVAPORATOR DATA					CONDENSER DATA					ELECTRICAL DATA			REF. TYPE	MANUFACTURER/ MODEL NO.	NOTES
			GPM	FOULING FACTOR	EWT (°F)	LWT (°F)	P.D. (FT.)	GPM	FOULING FACTOR	EWT (°F)	LWT (°F)	P.D. (FT.)	MCA	MOCP	VOLTS/PH/ HZ			
CH-3	CENTRIFUGUAL	1,250	2,985	0.0001	42.0	52.0	25.7	3,936	0.00025	85.0	93.9	36.4	1,272	2,000	460/3/60	R-514	TRANE/CVHF1300	ALL
NOTES: 1. REMOTE MOUNTED VARIABLE SPEED DRIVE. 2. 2 STAGE COMPRESSOR. 3. SINGLE POINT ELECTRICAL CONNECTION. 4. MARINE WATER BOX. 5. COMPLY WITH ASHRAE 90.1-2016. 6. PROVIDE FLOW SWITCH (FIELD INSTALLED). 7. TRANE SERVICE STARTUP. 8. 5 YEAR PART, LABOR, AND REFRIGERANT WARRANTY. 9. BACNET INTERFACE. 10. CHILLER HAS ALREADY BEEN PROCURED BY OWNER. CONTRACTOR TO INSTALL CHILLER. TRANE TO PROVIDE FACTORY STARTUP.																		



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MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: JH
CHECKED BY: TF

MECHANICAL
SCHEDULES &
DETAILS

M9.01

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FORT WORTH, TEXAS 76104

ABBREVIATIONS			
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
Ø	PHASE	IMC	INTERMEDIATE METAL CONDUIT
A	AMPS	IG	ISOLATED GROUND
AC	ALTERNATING CURRENT/ABOVE COUNTER	INCAND	INCANDESCENT
A.D.	AREA DRAIN	INSUL	INSULATION
ADJ	ADJACENT	INT	INTERCOM
AFF	ABOVE FINISHED FLOOR	J-BOX	JUNCTION BOX
AFG	ABOVE FINISHED GRADE	KVA	KILOVOLT AMPERES
ANSI	AMERICAN NATIONAL STANDARDS INSTITUTE	KW	KILOWATT
ARCH	ARCHITECTURAL	KWH	KILOWATT HOURS
ASTM	AMERICAN SOCIETY FOR TESTING MATERIALS	LBS	POUNDS
ASYM	ASYMMETRICAL	LOC.	LOCATED
ATS	AUTOMATIC TRANSFER SWITCH	LP	LIGHTING PANEL
AUX	AUXILIARY	LTG	LIGHTING
A/V	AUDIO/VISUAL	MA	MILLIAMPS
AWG	AMERICAN WIRE GAUGE	MACH	MACHINE
BLDG	BUILDING	MC	METAL-CLAD CABLE
B.F.P.	BACKFLOW PREVENTER	MCB	MAIN CIRCUIT BREAKER
B.V.	BALANCING VALVE	MCC	MOTOR CONTROL CENTER
B.D.	BALCONY DRAIN	MCM	THOUSAND CIRCULAR MILS
B.S.	BAR SINK	MCP	MOTOR CONTROL PANEL
B.T.	BATHTUB/SHOWER	MIC	MICROPHONE
BLW.	BELOW	MLO	MAIN LUGS ONLY
C	CONDUIT	MTD	MOUNTED

CONTINUATION ABBREVIATIONS			
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
C/B	CIRCUIT BREAKER	MTR	MOTOR
CAT	CATALOG	N	NEUTRAL
CKT	CIRCUIT	NC	NORMALLY CLOSED
CL	CLOCK	NEC	NATIONAL ELECTRICAL CODE
CLG	CEILING	NEMA	NATIONAL ELEC MANUFACTURERS ASSOC.
COL	COLUMN	NETA	NATIONAL ELECTRICAL TESTING ASSOC.
CONC	CONCRETE	NF	NON-FUSE
CONT	CONTINUOUS	NFPA	NATIONAL FIRE PROTECTION ASSOC.
CONV	CONVENIENCE	NL	ON NIGHT LIGHTING CIRCUIT
C/T	CURRENT TRANSFORMER	NO	NORMALLY OPEN
C.V.	CHECK VALVE	O/C	OVERCURRENT
C.O.	CLEANOUT	OL	OVERLOAD
C.W.	COLD WATER	LB.	POUNDS
COND.	CONDENSATE	P.S.I.	POUNDS PER SQUARE INCH
CONT.	CONTINUED	P	POLES
D.D.	DECK DRAIN	PB	PULL BOX
*F	DEGREES FAHRENHEIT	PF	POWER FACTOR
D.	DRAIN	PH	PHASE
D.F.U.	DRAINAGE FIXTURE UNITS	PLB	PLUMBING
DC	DIRECT CURRENT	PNL	PANEL
DIA	DIAMETER	PP	POWER PANEL
DP	DISTRIBUTION PANEL	P/T	POTENTIAL TRANSFORMER
DWG	DRAWING	PVC	POLYVINYL CHLORIDE
E	EXISTING TO REMAIN	PWR	POWER
EC	EMPTY CONDUIT	QUAD	QUADRUPLEX
EL	ELEVATION	R	REMOVE
ELEC	ELECTRIC	RC	REMOTE CONTROL
ELEV	ELEVATOR	RECEPT	RECEPTACLE
EM	ON EMERGENCY CIRCUIT	REL	RELOCATE
EQ	EQUIVALENT	REQ	REQUIRED
EMT	ELECTRICAL METALLIC TUBING	REV	REVERSE
EPO	EMERGENCY POWER OFF	RM	ROOM
EOMT	EQUIPMENT	RMS	ROOT MEAN SQUARE
(E)	EXISTING	SP	SPARE
F.D.C.	FIRE DEPARTMENT CONNECTION	SPD	SURGE PROTECTION DEVICE
F.H.C.	FIRE HOSE CABINET	SPECS	SPECIFICATIONS
F.	FIRE	SPKLR	SPRINKLER
FA	FIRE ALARM	SPKR	SPEAKER
FC	FOOT CANDLE	SO	SQUARE
FDR	FEEDER	STD	STANDARD
FIXT	FIXTURE	SURF	SURFACE
FL	FLUSH	SW	SWITCH
FLA	FULL LOAD AMPERES	SWBD	SWITCHBOARD
FLR	FLOOR	SWGR	SWITCHGEAR
FLUOR	FLUORESCENT	SUSP	SUSPENDED
FT	FEET	SYM	SYMMETRICAL
F.F.	FINISHED FLOOR	TEL	TELEPHONE
FLR.	FLOOR	TELCO	TELEPHONE COMPANY
F.C.O.	FLOOR CLEAN OUT	TV	TELEVISION
F.D.	FLOOR DRAIN	TYP	TYPICAL
F.S.	FLOOR SINK	UC	UNDERCOUNTER
G.P.H.	GALLONS PER HOUR	UFD	UNDERFLOOR DUCT
G.P.M.	GALLONS PER MINUTE	UL	UNDERWRITERS LABORATORIES, INC.
G.V.	GREASE VENT	V.T.R.	VENT THRU ROOF
G.C.O.	GRADE CLEAN OUT	V	VOLTS
GA	GUAGE	VA	VOLT AMPERES
GALV	GALVANIZED	VENT	VENTILATING
GFI	GROUND FAULT CIRCUIT INTERRUPTER	VERT	VERTICAL
GFP	GROUND FAULT PROTECTION	VP	VAPORPROOF
GND	GROUND	W.C.O.	WALL CLEAN OUT
GRC	GALVANIZED RIGID CONDUIT	W.H.	WALL HYDRANT
H.W.	HOT WATER	W.B.	WASHER BOX
H.W.R.	HOT WATER RETURN	W.C.	WATER CLOSET
HGT	HEIGHT	WG	WIRE GUARD
HOA	HAND OFF AUTOMATIC	WP	WEATHERPROOF
HORIZ	HORIZONTAL	XFMR	TRANSFORMER
HP	HORSEPOWER	YD	YARD
HTG	HEATING		
HVAC	HEATING, VENTILATING & AIR CONDITIONING	1/C, 2/C,	1 CONDUCTOR, 2 CONDUCTORS, ETC.
INV.	INVERT		
IEEE	INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS		

ELECTRICAL SYMBOLS LEGEND			
	EMERGENCY BATTERY PACK FIXTURE		TRANSFORMER
	CEILING EXIT LIGHT - DIRECTIONAL ARROWS WHERE INDICATED, SHADED QUADRANTS INDICATE ILLUMINATED FACES		400A / 3P AUTOMATIC TRANSFER SWITCH AMPS / POLES AS INDICATED ON PLANS
	WALL EXIT LIGHT - DIRECTIONAL ARROWS WHERE INDICATED SHADED QUADRANTS INDICATE ILLUMINATED FACES		MOTOR
	LIGHTING CONTROL POWER PACK		MAGNETIC MOTOR STARTER, H-O-A-P U.O.N.
	OCCUPANCY SENSOR CEILING MOUNT		COMBINATION MOTOR CONTROLLER AND NON-FUSED DISCONNECT SWITCH
	VACANCY SENSOR CEILING MOUNT		VARIABLE FREQUENCY DRIVE FURNISHED BY DIV 23, CONNECTED BY DIV 26
	TOGGLE SWITCH, 15A / 277VAC UON 2 = DOUBLE POLE 3 = THREE-WAY 4 = FOUR-WAY a = CONTROLS OUTLET OR FIXTURE NOTED "a" K = KEY OPERATED P = WITH PILOT LIGHT T = INTERVAL TIMER, 30 MIN. UON		INFRARED SENSORS. SENSORS SHALL BE CONNECTED TO CORRIDOR LIGHTING
	KEY PAD, CONTRACTOR RESPONSIBLE TO PROVIDE ALL REQUIRED HARDWARE AND ELECTRICAL CONNECTIONS.		DOOR CONTACT, CONTRACTOR RESPONSIBLE TO PROVIDE ALL REQUIRED HARDWARE AND ELECTRICAL CONNECTIONS.
	MO = SPDT MOMENTARY CONTACT D = DIMMER SWITCH OC = OCCUPANCY SENSOR VS = VACANCY SENSOR LV = LOW VOLTAGE		PUSHBUTTON K = KEY OPERATED SWITCH (U/OFF/DOWN POSITION) E = EMERGENCY POWER OFF BUTTON RELAY
	PHOTOCELL CONTROL SWITCH		100 UNFUSED SWITCH - 100 AMP 3 POLE, UON
	TIME CLOCK		100 CIRCUIT BREAKER - 100 AMP FRAME, 3 POLE, UON
	DUPLEX CONVENIENCE RECEPTACLE NUMERAL INDICATES CKT NUMBER IG = ISOLATED GROUND (ORANGE DEVICE) G = GROUND FAULT INTERRUPTER 20A = 20 AMP RATED RECEPTACLE * * ALL RECEPTACLES ON DEDICATED 20 AMP CIRCUITS SHALL BE RATED 20A		100/100 FUSED SWITCH 100 AMP SWITCH/100 AMP TYPE "FA" FUSE 3 POLE UON
	QUADRUPLX CONTROL SWITCH		FACFP FIRE ALARM CONTROL PANEL
	SINGLE POWER RECEPTACLE, 20A, 125V 3W GNDG UON.		FAAP FIRE ALARM REMOTE ANNUNCIATOR
	FLUSH FLOOR DUPLEX RECEPTACLE		FAS FIRE ALARM STROBE
	FLUSH FLOOR QUADRUPLEX RECEPTACLE		FASP FIRE ALARM SPEAKER STROBE
	PLUGMOLD WITH DUPLEX OUTLETS, 18" OC U.O.N. "12" INDICATES OUTLETS 12" OC "24" INDICATES OUTLETS 24" OC. "C" INDICATES WITH COMM COMPARTMENT		FAS FIRE ALARM SPEAKER
	TELEVISION SIGNAL OUTLET		FATS FIRE ALARM TAMPER SWITCH
	WALL MOUNTED DATA/TELEPHONE		FAWF FIRE ALARM WATER FLOW
	FLOOR DATA OUTLET		S SMOKE DETECTOR
	WALL TELEPHONE OUTLET WITH PLATE (W-WALL MOUNTED)		D DUCT SMOKE DETECTOR
	CEILING MOUNTED JUNCTION BOX		H HEAT DETECTOR
	FLUSH WALL MOUNTED JUNCTION BOX FOR MOUNTING HEIGHT REFER TO ARCHITECTURAL AND MEDICAL EQUIPMENT PLAN DRAWINGS.		F MANUAL PULL STATION
	FLOOR MOUNTED JUNCTION BOX		DH FIRE ALARM MAGNETIC DOOR HOLD-OPEN
	CONDUIT TURNING UP/DOWN		F FIREMAN'S PHONE
	CAPPED CONDUIT		FT FIREMAN'S PHONE HANDSET
	GROUND CONNECTION		GAP GENERATOR ANNUNCIATOR PANEL
	CONDUIT HOME RUN TO PANEL BOARD "1LB" WITH CIRCUITS 1,3,5		NCM NURSE CALL MASTER STATION
	DISCONNECT SWITCH - 30A, NON-FUSED, 3-POLE UON. NEMA 1 ENCLOSURE EXCEPT "3R" INDICATES NEMA 3R.		P NURSE CALL PATIENT STATION
	FLUSH MOUNTED CIRCUIT BREAKER 100A FRAME/60A TRIP 3 POLE UON		SD NURSE CALL STAFF/DUTY STATION
	PANELBOARD		CB NURSE CALL CODE BLUE
	LIGHTING RELAY PANEL		Z CEILING MOUNTED ZONE LIGHT
			D CEILING MOUNTED DOME LIGHT
			NCC NURSE CALL CENTRAL EQUIPMENT CABINET

INDEX TO ELECTRICAL DRAWINGS	
E0.00	ELECTRICAL INFORMATION SHEET
E1.01A	ENLARGED FLOOR PLAN - CHILLER PLANT AREA A - ELECTRICAL DEMOLITION
E1.01B	ENLARGED FLOOR PLAN - CHILLER PLANT AREA B - ELECTRICAL DEMOLITION
E2.01A	ENLARGED FLOOR PLAN - CHILLER PLANT AREA A - ELECTRICAL
E2.01B	ENLARGED FLOOR PLAN - CHILLER PLANT AREA B - ELECTRICAL
E4.00	ELECTRICAL RISER DIAGRAM
E5.00	ELECTRICAL DETAILS

ISSUE FOR CONSTRUCTION 03/05/21
NO. ISSUED FOR: DATE:



03/05/21

MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: CD
CHECKED BY: MW

ELECTRICAL NOTES,
LEGENDS, AND SYMBOLS

E0.00

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1500 S. Main Street
FORT WORTH, TEXAS 76104

GENERAL NOTES

- ALL WORK SHALL COMPLY WITH ALL PERTINENT NATIONAL, STATE, AND LOCAL ORDINANCES AND CODES, AND ALL AMERICAN DISABILITIES ACT (ADA) REQUIREMENTS.
 - DRAWINGS ARE DIAGRAMMATIC; COORDINATE THE ROUGH-IN LOCATION, CONNECTION TYPE AND TERMINATION REQUIREMENTS WITH EQUIPMENT INSTALLERS PRIOR TO ROUGH-IN.
 - REFER TO ARCHITECTURAL DRAWINGS FOR EXACT PROJECT PHASING REQUIREMENTS.
 - REMOVE ALL ELECTRICAL AND SPECIAL SYSTEMS DEVICES AND EQUIPMENT (LIGHT FIXTURES, FIRE ALARM DEVICES, WIRING DEVICES, SPEAKERS, ETC.) SHOWN DASHED AND WITHIN DEMOLITION AREA INDICATED BY DASHED WALLS, UNO.
- FOR ALL REMOVED WIRING DEVICES, REMOVE ASSOCIATED BRANCH CIRCUITING AND DATA/COMMUNICATION WIRING AND CONDUIT FROM DEVICE TO SOURCE, UNLESS NOTED OTHERWISE OR UNLESS REQUIRED TO MAINTAIN SERVICE TO DEVICES IN ADJACENT AREAS.
- MAINTAIN SERVICE TO LIGHT FIXTURES, WIRING DEVICES, FIRE ALARM AND SPECIAL SYSTEM DEVICES OUTSIDE OF WORK AREA.
 - "R" DENOTES REMOVE AND REINSTALL TO LOCATION SHOWN ON SHEET 1/E2.01A & 1/E1.01B.

KEYED NOTES

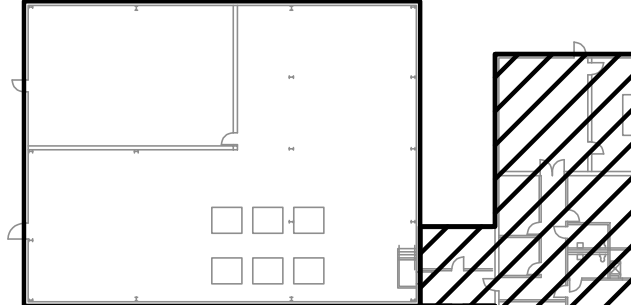
(NOT ALL KEYED NOTES APPEAR ON THIS SHEET.)

- BOILER 1 & 2 VFD'S TO BE REMOVED AND REINSTALLED. RE: 1/E2.01.
- EXISTING CHP-2 & 3 VFD'S TO BE REMOVED. RE: 1/E2.01.
- KEYED SWITCH AND TOGGLE SWITCH TO BE REMOVED AND REINSTALLED. RE: 1/E2.01.
- CHILLED WATER PUMP #2 FLOW METER TO BE REMOVED AND REINSTALLED. RE: 1/E2.01.
- REMOVE EXISTING METAL HALIDE LIGHT FIXTURE. RETAIN BRANCH CIRCUIT TO SERVE NEW LIGHT FIXTURES. RE: 1/E2.01.
- INTERCEPT INCOMING 3000A FEEDER SERVING TSP-MCC.

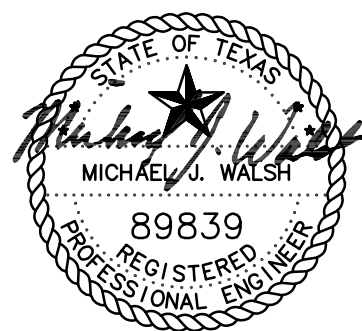
LINE TYPE LEGEND

---	DEMOLISH
---	EXISTING
---	NEW

KEY PLAN



ISSUE FOR CONSTRUCTION 03/05/21
NO. ISSUED FOR: DATE:



03/05/21

MAIN CENTRAL PLANT CHILLER ADDITION

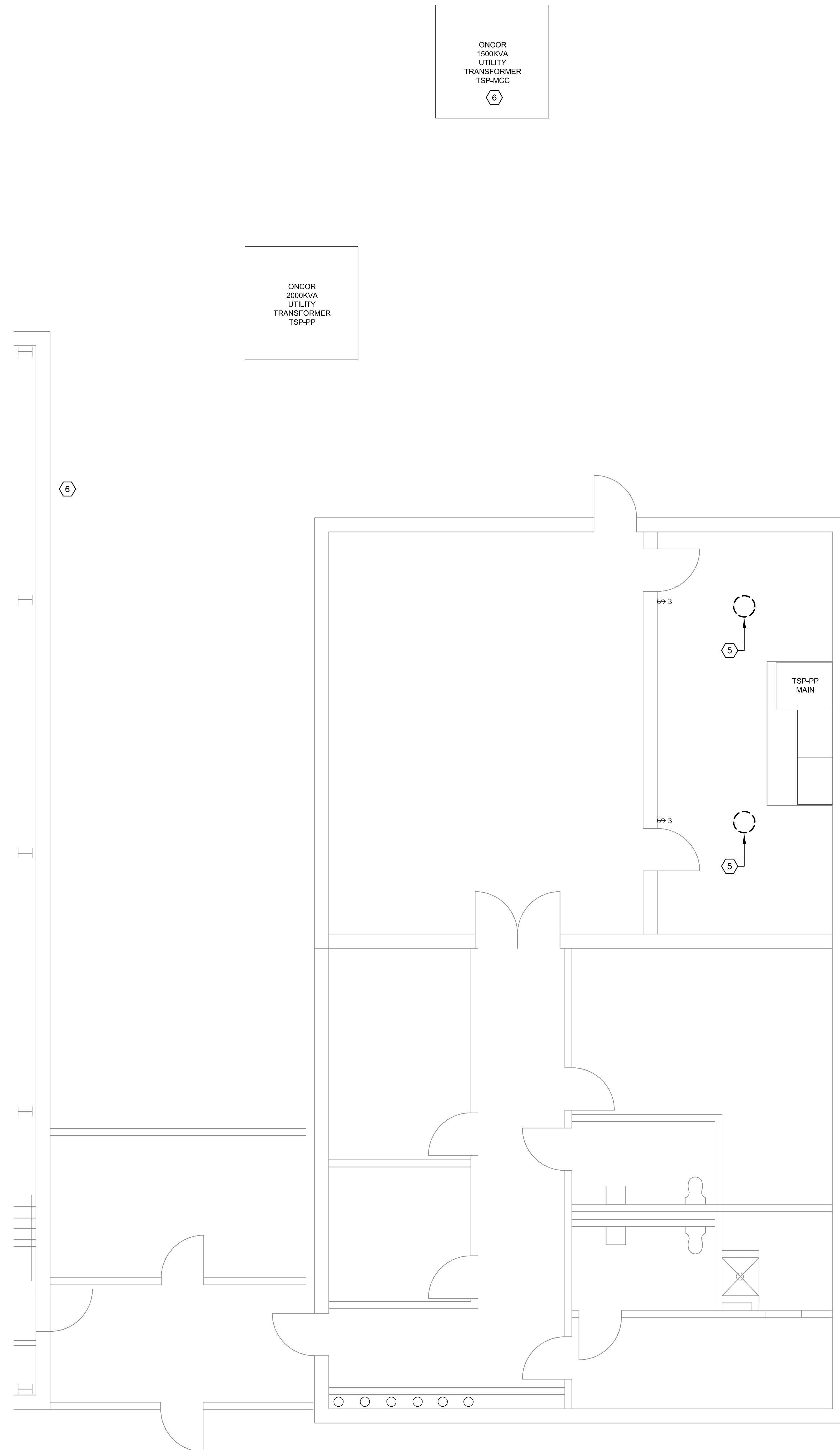
SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: CD
CHECKED BY: MW

ENLARGED CHILLER PLANT AREA A - ELECTRICAL DEMOLITION

E1.01A

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1500 S. Main Street
FORT WORTH, TEXAS 76104



GENERAL NOTES

1. ALL WORK SHALL COMPLY WITH ALL PERTINENT NATIONAL, STATE, AND LOCAL ORDINANCES AND CODES, AND ALL AMERICAN DISABILITIES ACT (ADA) REQUIREMENTS.
 2. DRAWINGS ARE DIAGRAMMATIC; COORDINATE THE ROUGH-IN LOCATION, CONNECTION TYPE AND TERMINATION REQUIREMENTS WITH EQUIPMENT INSTALLERS PRIOR TO ROUGH-IN.
 3. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT PROJECT PHASING REQUIREMENTS.
 4. REMOVE ALL ELECTRICAL AND SPECIAL SYSTEMS DEVICES AND EQUIPMENT (LIGHT FIXTURES, FIRE ALARM DEVICES, WIRING DEVICES, SPEAKERS, ETC.) SHOWN DASHED AND WITHIN DEMOLITION AREA INDICATED BY DASHED WALLS, UNO.
- FOR ALL REMOVE WIRING DEVICES, REMOVE ASSOCIATED BRANCH CIRCUITING AND DATA/COMMUNICATION WIRING AND CONDUIT FROM DEVICE TO SOURCE UNLESS NOTED OTHERWISE OR UNLESS REQUIRED TO MAINTAIN SERVICE TO DEVICES IN ADJACENT AREAS.
5. MAINTAIN SERVICE TO LIGHT FIXTURES, WIRING DEVICES, FIRE ALARM AND SPECIAL SYSTEM DEVICES OUTSIDE OF WORK AREA.
6. "R" DENOTES REMOVE AND REINSTALL TO LOCATION SHOWN ON SHEET/E/01A & 1/E/01B.

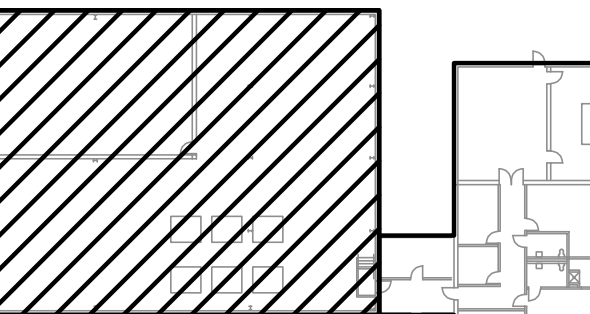
KEYED NOTES

(NOT ALL KEYED NOTES APPEAR ON THIS SHEET.)

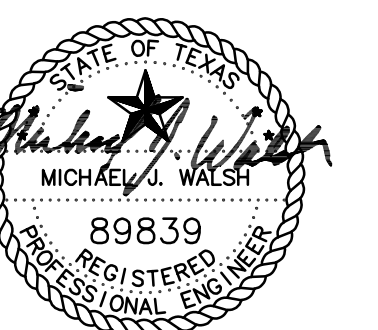
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LINETYPE LEGEND	
	DEMOLISH
	EXISTING
	NEW

KEY PLAN



ISSUE FOR CONSTRUCTION	03-05-21
ISSUED FOR:	DATE:



3/05/21

MAIN CENTRAL PLANT CHILLER ADDITION

SCALE:	1/4"=1'-0"
PROJECT NO:	121020.00
DRAWN BY:	CD
CHECKED BY:	MW

ENLARGED CHILLER PLANT
AREA B - ELECTRICAL
DEMOLITION

E1.01B © 2021

GENERAL NOTES

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- REFER TO ELECTRICAL INFORMATION SHEET E0.01 FOR LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO ARCHITECT FOR EXACT LOCATIONS AND ELEVATIONS OF WIRING DEVICES.
- REFER TO PANELBOARD SCHEDULES FOR WIRE AND CONDUIT SIZES TO EQUIPMENT REQUIRING SPECIAL CONNECTIONS.
- DISCONNECT SWITCHES OR SPECIAL RECEPTACLE TYPES ARE TO BE IDENTIFIED MATCHING THE EQUIPMENT FOR WHICH IT SERVES.
- RUN #10 AWG TO FIRST DEVICE OF CIRCUIT IN HOMERUNS BEYOND 75' IN DISTANCE.
- ALL SINGLE-PHASE RECEPTACLES RATED 150 V TO GROUND OR LESS, 50 A OR LESS AND THREE-PHASE RECEPTACLES RATED AT 150V TO GROUND OR LESS, 100 A OR LESS INSTALLED IN (1) BATHROOMS, (2) KITCHENS, (3) ROOFTOPS, (4) OUTDOORS, (5) SINKS, (6) INDOOR WET LOCATIONS, (7) LOCKER ROOMS WITH ASSOCIATED SHOWERING FACILITIES, (8) GARAGES, SERVICE BAYS, AND SIMILAR AREAS OTHER THAN VEHICLE EXHIBITION HALLS AND SHOWROOMS, (9) CRAWL SPACES AT OR BELOW GRADE LEVEL.
- ALL TERMINALS IN ELECTRICAL EQUIPMENT IN THIS WORK SHALL BE U.L. LISTED FOR 75°C.
- ROUTING OF ALL UNDER FLOOR CONDUITS/ SHALL BE COORDINATED IN THE FIELD. CONTRACTOR SHALL CORE DRILL, PATCH AND PAINT AS REQUIRED, COORDINATE WITH ARCHITECT/OWNER FOR FINISHES.

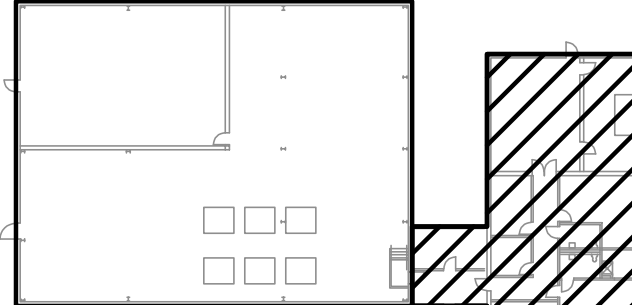
KEYED NOTES

- REINSTALL BOILER 1 & 2 VFD'S. EXTEND BRANCH CIRCUIT AS REQUIRED TO ACCOMMODATE NEW LOCATION. RE: 1/E1.01A FOR ORIGINAL LOCATION.
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- REFER TO SHEET E4.01 FOR NEW REQUIREMENT.
- REINSTALL KEYED SWITCH AND TOGGLE SWITCH. EXTEND CABLING AND CONDUIT AS REQUIRED TO ACCOMMODATE NEW LOCATION. RE: 1/E1.01A FOR ORIGINAL LOCATION.
- REINSTALL CHILLED WATER PUMP #2 FLOW METER TO LOCATION AS SHOWN. RE: 1/E1.01A FOR ORIGINAL LOCATION.
- PROVIDE LED 4' STRIP LIGHTS LITHONIA #ZLN-48-500LM-FST-4VOLT-40K-80CRIWH OR EQUAL. MAKE BRANCH CIRCUIT CONNECTION MADE AVAILABLE BY DEMOLITION. CHAIN MOUNT LIGHT FIXTURES AT 10'-0" AFF.

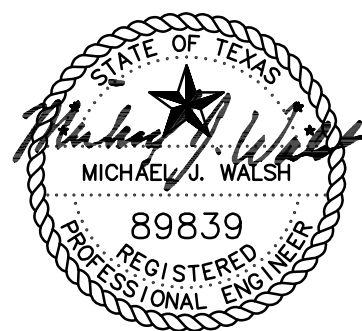
LINE TYPE LEGEND

---	DEMOLISH
---	EXISTING
---	NEW

KEY PLAN



ISSUE FOR CONSTRUCTION 03/05/21
NO. ISSUED FOR: DATE:



03/05/21

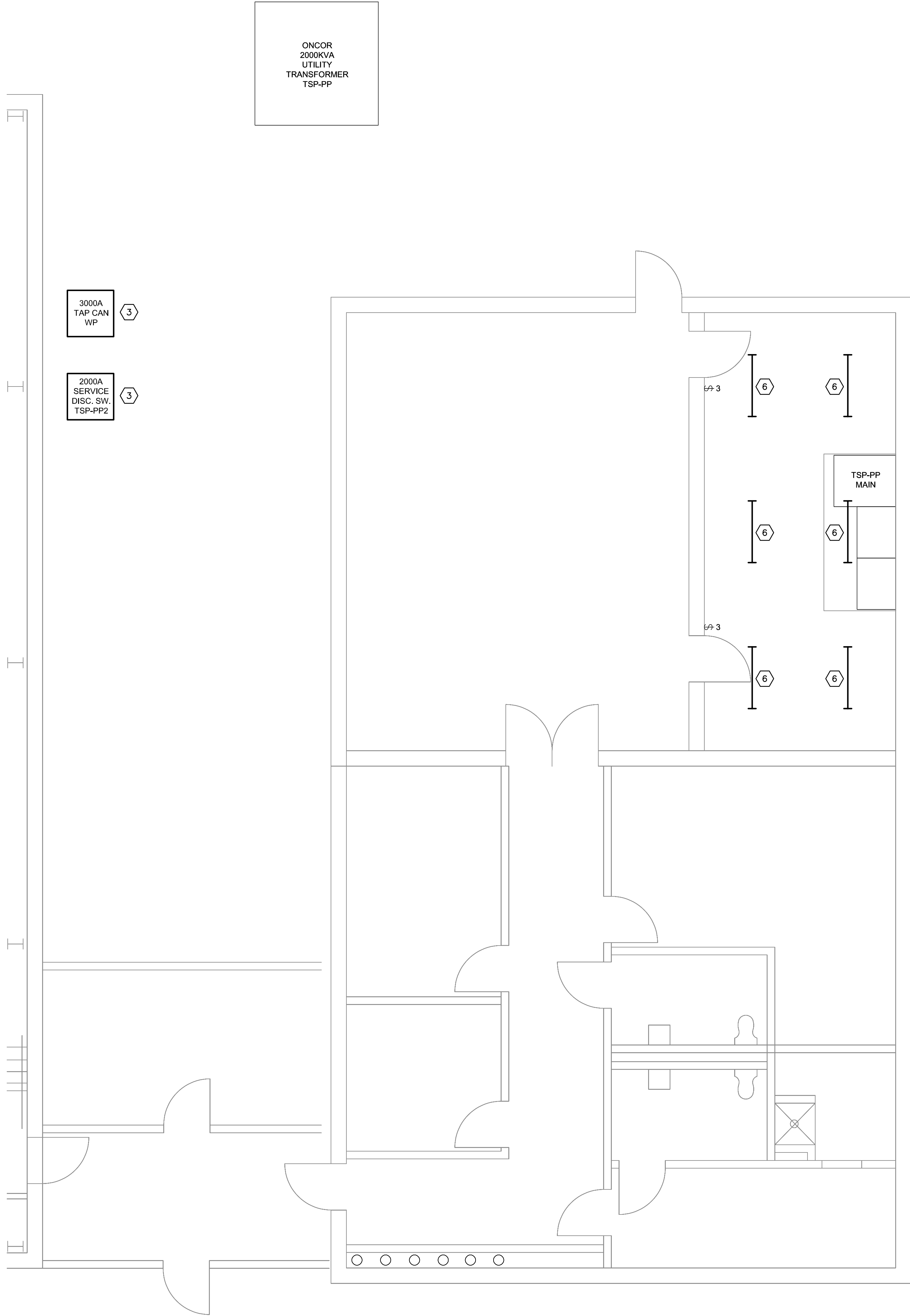
MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: CD
CHECKED BY: MW

ENLARGED CHILLER
PLANT AREA A -
ELECTRICAL

E2.01A
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1500 S. Main Street
FORT WORTH, TEXAS 76104



GENERAL NOTES

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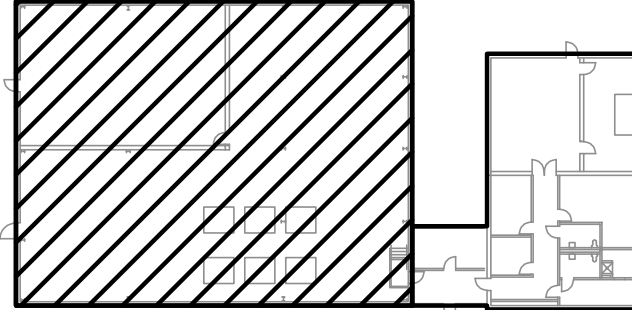
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LINE TYPE LEGEND

----	DEMOLISH
=====	EXISTING
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KEY PLAN



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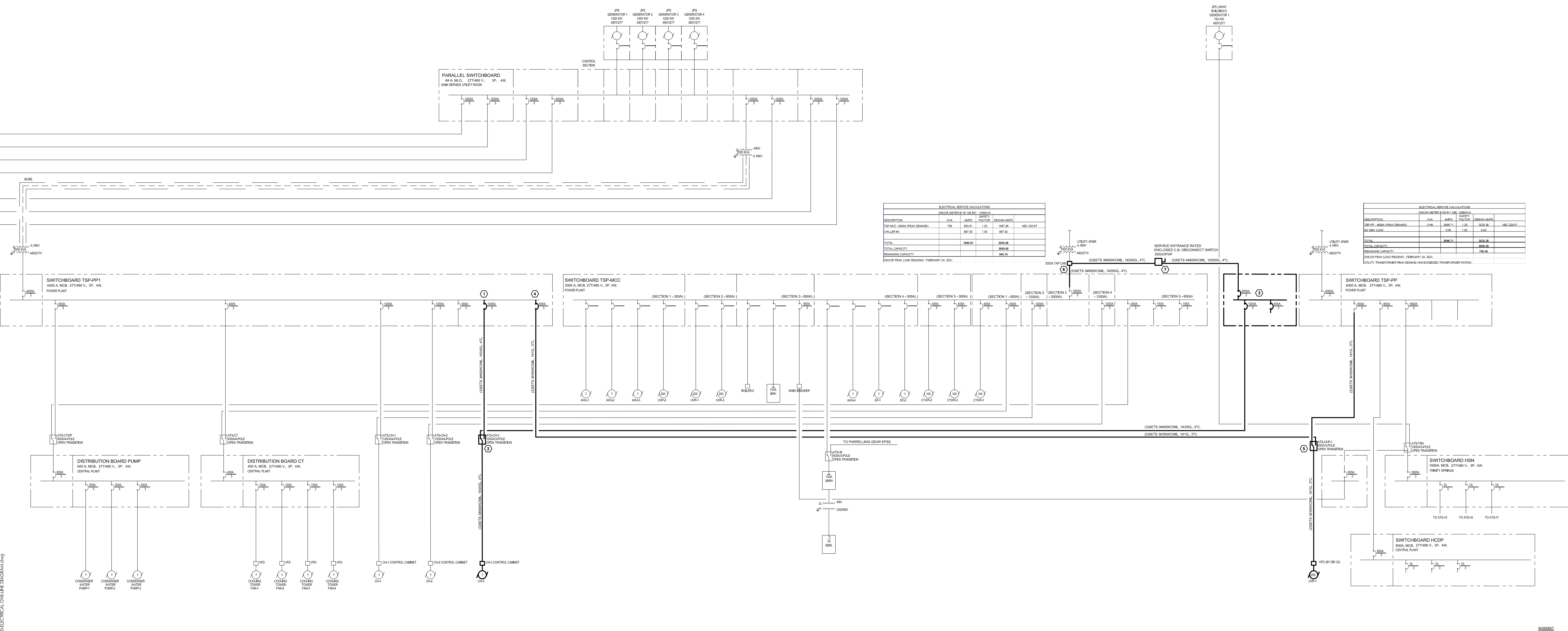
MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: CD
CHECKED BY: MW

ENLARGED CHILLER PLANT
AREA B – ELECTRICAL

E2.01B
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GENERAL NOTES	KEYED NOTES								
1. ALL WORK SHALL COMPLY WITH ALL PERTINENT NATIONAL, STATE, AND LOCAL ORDINANCES AND CODES, AND ALL AMERICAN DISABILITIES ACT (ADA) REQUIREMENTS. 2. DRAWINGS ARE DIAGRAMMATIC; COORDINATE THE ROUGH-IN LOCATION, CONNECTION TYPE AND TERMINATION REQUIREMENTS WITH EQUIPMENT INSTALLERS PRIOR TO ROUGH-IN. 3. REFER TO ELECTRICAL INFORMATION SHEET E0.00 FOR LEGEND, ABBREVIATIONS AND NOTES. 4. NEW EQUIPMENT TO MATCH EXISTING MANUFACTURERS INCLUDING CIRCUIT BREAKER TYPES AND AIC RATING FOR AT'SS, SWITCHBOARDS, MOTOR CONTROL CENTERS, ETC. 5. PROVIDE ANY 120V CONTROL POWER FROM PANELBOARD XBPL FOR MECHANICAL CONTROLS VALVES. MAKE CONNECTION TO NEW 20A/1P CIRCUIT BREAKER.	① PROVIDE NEW 1200A/3P LSIIG CIRCUIT BREAKER IN EXISTING SWITCHBOARD TSP-PP1. ROUTE CONDUIT UP AND THROUGH BRICK WALL TO SERVE ATS-CH3. ② PROVIDE NEW 1200A/4P AUTOMATIC TRANSFER SWITCH ATS-CH3. ROUTE FEEDER THROUGH CH-3 CONTROL CABINET/ADAPTIVE FREQUENCY DRIVE (AFD) AND TO CH-3. ③ PROVIDE NEW SWITCHBOARD TSP-PP2, 2000A MCB, 277/480V, 3-PHASE, 4-WIRE, 65K AIC. ④ PROVIDE NEW 600A/3P LSIIG CIRCUIT BREAKER IN EXISTING SWITCHBOARD TSP-PP1. ROUTE CONDUIT UP AND THROUGH BRICK WALL TO SERVE ATS-CHP-1. ⑤ PROVIDE NEW 600A/3P AUTOMATIC TRANSFER SWITCH ATS-CHP-1. ROUTE FEEDER THROUGH CHP-1 CONTROL VARIABLE FREQUENCY DRIVE (VFD) AND TO CHP-1. ⑥ PROVIDE NEW 3000A TAP CAN, WP. ⑦ PROVIDE NEW 2000A MCB, WP ENCLOSED C.B. DISCONNECT SWITCH.								
<table><tr><th colspan="2">LINETYPE LEGEND</th></tr><tr><td>---</td><td>DEMOLISH</td></tr><tr><td>---</td><td>EXISTING</td></tr><tr><td>---</td><td>NEW</td></tr></table>		LINETYPE LEGEND		---	DEMOLISH	---	EXISTING	---	NEW
LINETYPE LEGEND									
---	DEMOLISH								
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---	NEW								



ISSUE FOR CONSTRUCTION 03-05-21
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MAIN CENTRAL PLANT
CHILLER ADDITION

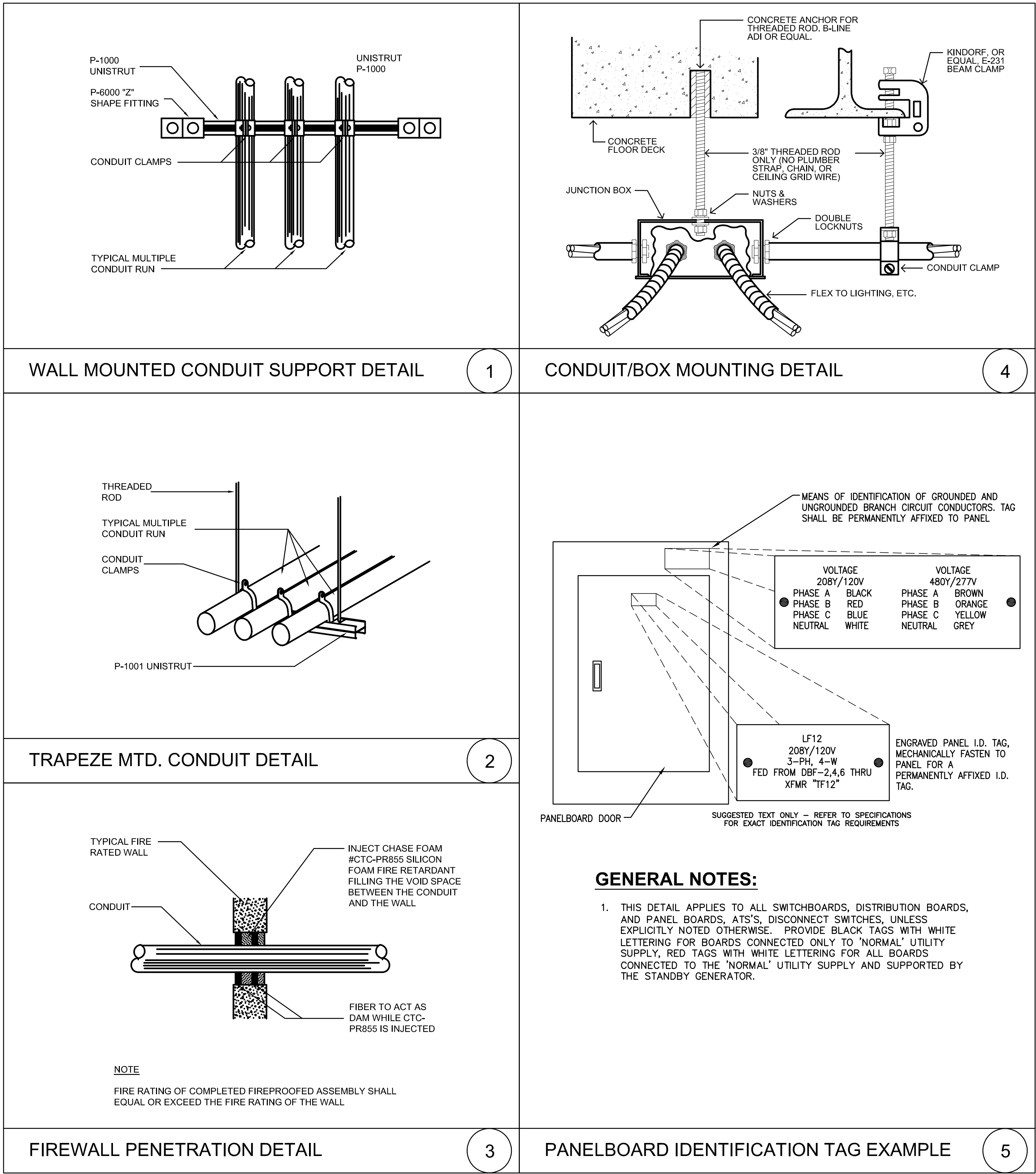
SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: CD
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ELECTRICAL
ONE-LINE
DIAGRAM

E4.00

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1500 S. Main Street
FORT WORTH, TEXAS 76104



ISSUE FOR CONSTRUCTION 03-05-21
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03/05/21

MAIN CENTRAL PLANT
CHILLER ADDITION

SCALE: 1/4"=1'-0"
PROJECT NO: 121020.00
DRAWN BY: CD
CHECKED BY: MW

ELECTRICAL DETAILS

E5.00

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